

# Information and Communication Technology

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Lecture : 01

Department of Computer Science  
GPGC SWABI (Khyber Pakhtunkhwa)

Fall 2025 Semester  
BS Course: ICT

Course Instructor :  
Mr. Ihtisham Ul Haq  
Lecturer (CS)  
GPGC Swabi

# Objectives

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- At the end of course the students will be able to:
  - Understand fundamental ICT concepts and terminology.
  - Develop proficiency in basic computer networking concepts.
  - Explore the impact of ICT on society.
  - Learn to use MS Office for different tasks.
  - Apply ethical and responsible practices in information and communication technology.

# References

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- ❑ *Introduction to Computers 6th International Edition, Peter, N. McGraw-Hill*
- ❑ *Using Information Technology: A Practical Introduction to Computer & Communications, 6th Edition. Williams, S. McGraw-Hills.*
- ❑ *Fundamentals of Information Technology, Alexis L Mathewsleon Leon Press.*
- ❑ *ICT IT series*

# Subject Marks Breakdown

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- ❑ Midterm Marks : 20
- ❑ Internal Marks : 15
  - Presentation, Quiz, Assignment : 10
  - Attendance : 05
- ❑ Viva Marks : 30
- ❑ Final Term Marks : 35

# Motivation Byte

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# Contents

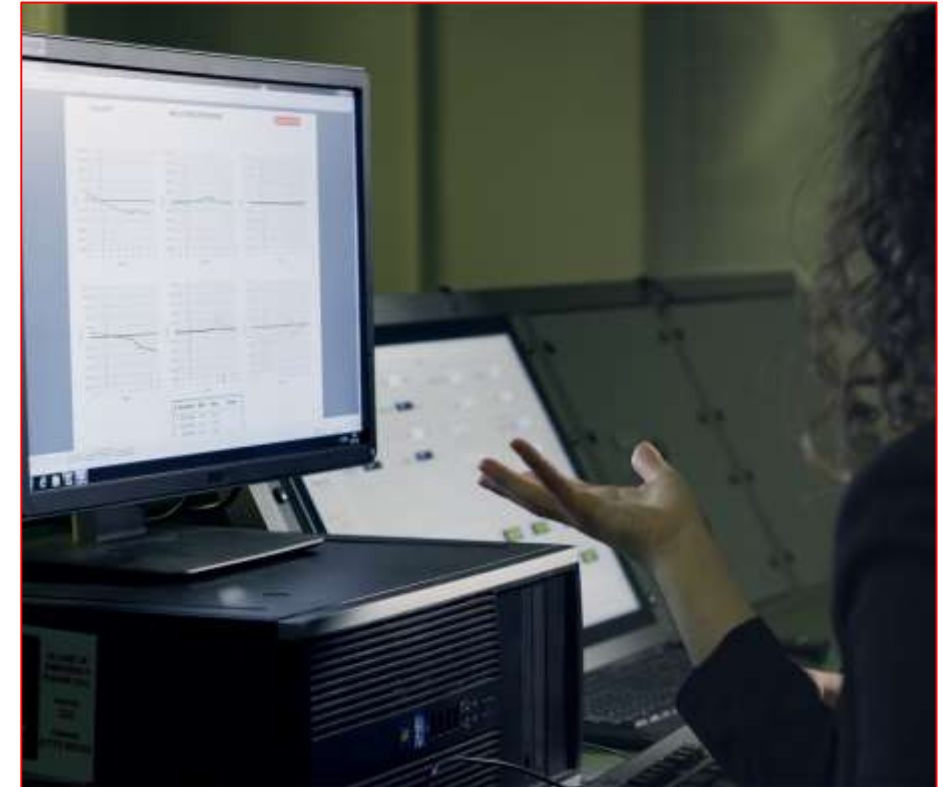
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- ☐ Introduction to Computer
  - ☐ Computer
  - ☐ Data Vs Information
  - ☐ Why we Use Computers
- ☐ Applications of Computers
- ☐ History of Computers

# Introduction to Computer

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- ❑ Computer Definition:
  - “The computer is an *electronic machine* that takes data as input, process it, store results in memory, and displays it.”
- ❑ It takes the data as an input and processes that data to perform tasks.
- ❑ A computer follows instructions to get things done quickly and accurately.
- ❑ Computer has memory to store information so that it can quickly access it when needed.



# Introduction to Computer (Cont.)

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- ❑ Computer is a device that transforms data into meaningful information.
- ❑ It processes the input according to the set of instructions and gives the desired output.
- ❑ A Computer can perform the following set of functions:
  - Accept data
  - Store data
  - Process data as desired
  - Retrieve the stored data as and when required
  - Output the result in desired format.

## Did You Know?

The word computer was originally coined as an acronym which stands for **C**ommonly **O**perating **M**achine **P**articularly **U**sed for **T**echnology, **E**ducation, and **R**esearch.



# Introduction to Computer (Cont.)

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## ☐ **Data Vs. Information**

- ☐ **Data:** It is the raw facts and figures.
- ☐ Data is fed into the computer along with a set of instructions.
- ☐ **Information:** Data represented in useful and meaningful form is information.
- ☐ Information is the output of the computer.

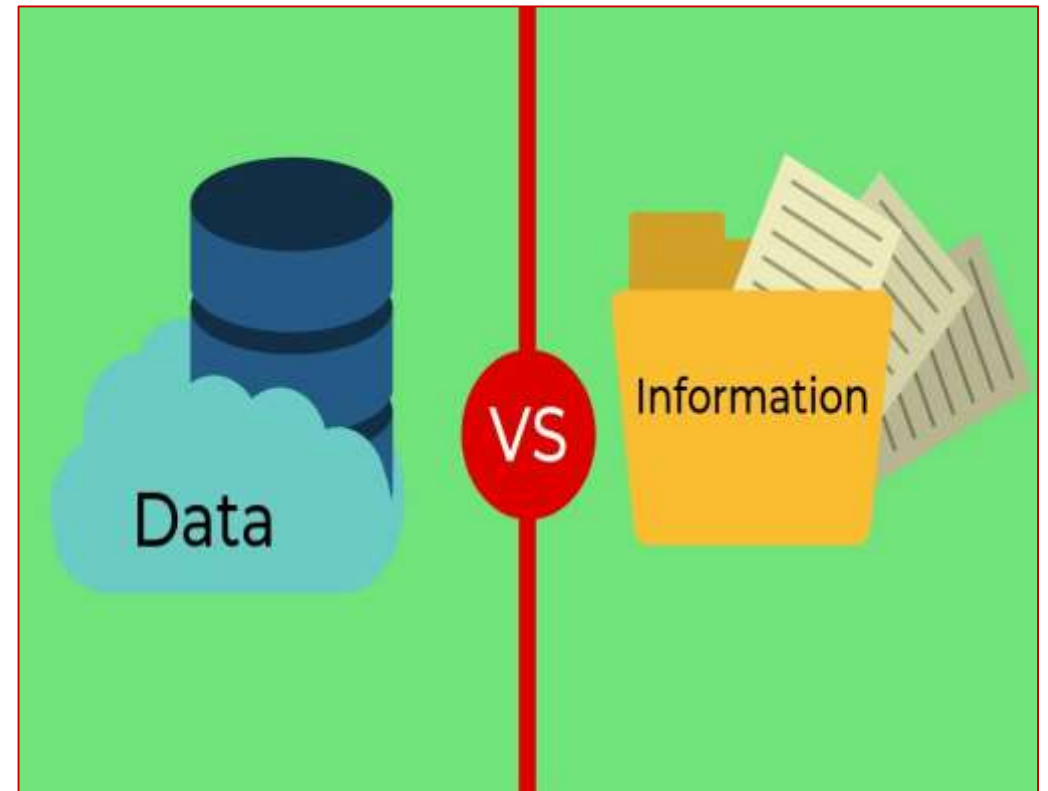
# Introduction to Computer (Cont.)

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## ❑ Data Vs. Information

### ❑ Examples from real life?

- Weather (Temperature, Humidity, Wind, atmospheric pressure vs Sunny/Rainy/Snow)
- Student Grades (Individual scores in papers vs report card)
- Census Population Data (Population, age, gender, location vs population growth, age distribution)



# Introduction to Computer (Cont.)

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## ☐ **Why we Use Computers?**

- ☐ Speed
- ☐ Accuracy
- ☐ High Storage
- ☐ Diligence
- ☐ Versatile
- ☐ Reliable
- ☐ Automation



# Applications of Computers

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- ❑ Computers have a wide range of applications across various industries and in our daily lives.
- ❑ Here are some of the most common and important applications of computers:
- ❑ **Personal Computing:** Personal computers are used for tasks such as word processing, web browsing, email communication, and entertainment.
- ❑ **Business and Finance:** Computers are essential for businesses, including accounting, inventory management, etc. They are also used for online banking and trading.
- ❑ **Education:** Computers are used in schools and universities for research, online learning, and administrative purposes. Educational software and interactive multimedia tools are widely used in the classroom.
- ❑ **Healthcare:** Computers are used for patient records management, diagnostic imaging like MRI, CT scans, etc.

# Applications of Computers

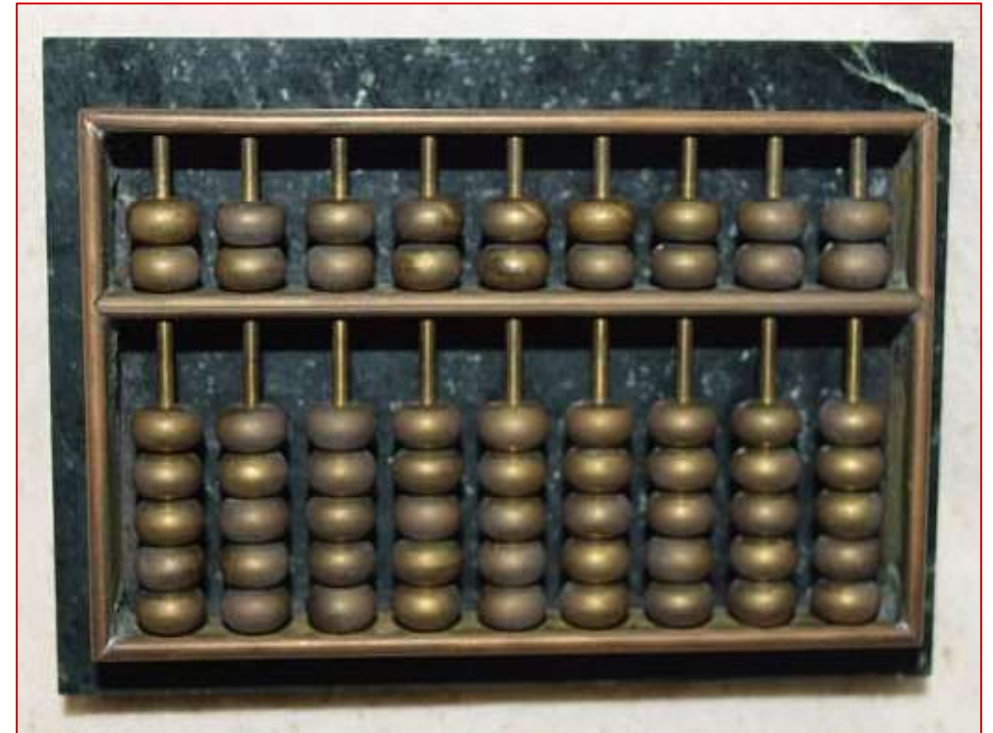
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- ❑ **Communication:** Computers power the internet and telecommunications networks, enabling email, video conferencing, social media, and instant messaging.
- ❑ **E-commerce:** Online shopping, payment processing, and inventory management are all powered by computers.
- ❑ **Weather Forecasting:** Supercomputers are used to process vast amounts of weather data for accurate weather predictions.
- ❑ **Gaming:** Video game consoles and gaming PCs are used for gaming.

# History of Computers

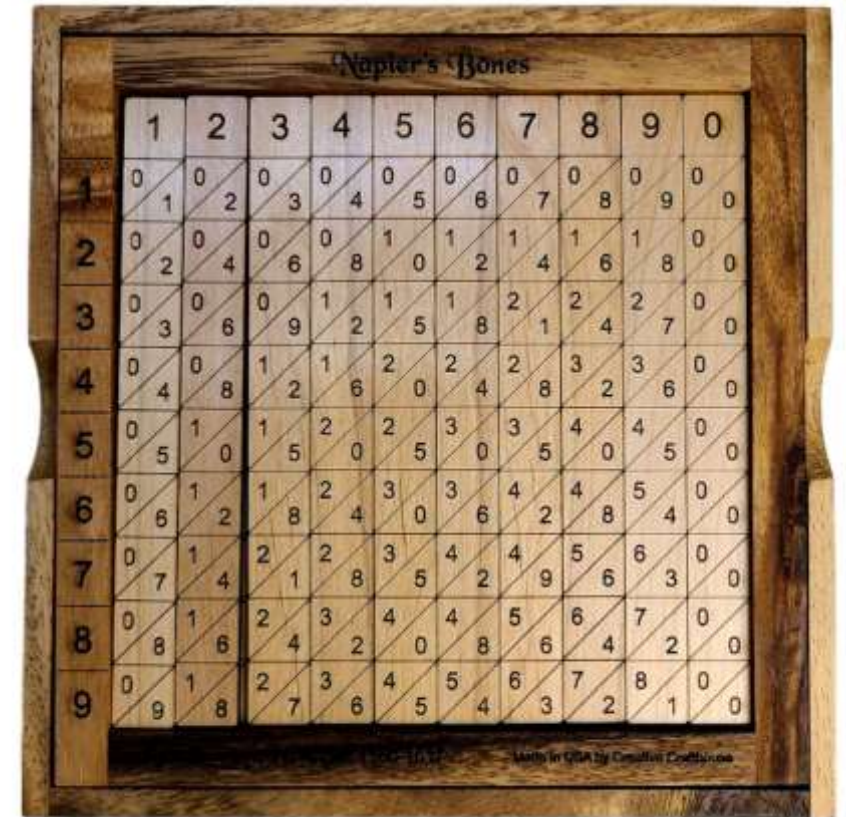
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- ❑ Computer has its roots in history dating back to 3000 B.C.
- ❑ The earliest known counting device known as Abacus was known to exist 3000 B.C.
- ❑ It was built out of wood and beads.
- ❑ It was used to perform addition, subtraction, multiplication, and division.



# History of Computers (Cont.)

- ❑ In early 1600, Jhon Napier invented calculating rods, called Napier's Bone.
- ❑ Napier's Bone allowed quick multiplication of large numbers accurately.



# History of Computers (Cont.)

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- ❑ One of the first mechanical calculator, Pascaline, was developed by Blaise Pascal in 1642.





# History of Computers (Cont.)

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- ❑ Charles Babbage, known as the “Father of Modern Digital Computer” developed a calculator called “Difference Engine”.
- ❑ The Difference engine was invented in 1821.
- ❑ He also planned to build “Analytical Engine”, but due to technological limitations could not built it.



# History of Computers (Cont.)

- ❑ In WWII, scientists were researching to make a device that can decipher enemy communications.
- ❑ Enigma Machine?
- ❑ Alan Turing of the British was successful in decoding the Nazi's messages.
- ❑ Alan Turing is also called the father of Computer Science.



**Alan Turing's Bombe**

# Extra...

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- ❑ **Artificial Learning Based Tools**
  - ❑ Chat GPT

# Any Question?

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# Thank you

# Information and Communication Technology

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Lecture : 02

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- ❑ Generations of Computers
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  - ❑ 2<sup>nd</sup> Generation Computers
  - ❑ 3<sup>rd</sup> Generation Computers
  - ❑ 4<sup>th</sup> Generation Computers
  - ❑ 5<sup>th</sup> Generation Computers

# Generation of Computers

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- ❑ Computer generations refer to the advancement in computer technology over time.
- ❑ With each new generation, the circuits becomes smaller and more advanced than the previous one.
- ❑ As a result the efficiency, speed, power, and memory of the computers have increased.
- ❑ There are five generations of computers.

# Generation of Computers (Cont.)

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## ❑ The First Generation: 1940-1956

- ❑ Main Component : Vacuum Tubes
- ❑ The first generation computers were slow, expensive, and often unreliable.
- ❑ These computers took lots of space spanning many rooms and produces a lot of heat.
- ❑ These computers used Machine language for instructions.
- ❑ Examples include: **ENIAC** (Electronic Numerical Integrator and Calculator), **UNIVAC** (Universal Automatic Computer)





# Generation of Computers (Cont.)

## ❑ Characteristics of First Generation

- ❑ Used Vacuum tubes
- ❑ Were large and expensive
- ❑ Limited memory storage
- ❑ Low processing speed

### Did You Know?

The term bug in computer refers to any flaw/error that causes computer to stop functioning normally.



**ENIAC**

# Generation of Computers (Cont.)

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## ❑ The Second Generation: 1956-1964

- ❑ Main Component : Transistors
- ❑ Transistors replaced vacuum tubes in the second generation of computers.
- ❑ Transistor was invented in 1947 at Bell Labs.
- ❑ The transistor was faster, more reliable, smaller, and much cheaper to build than a vacuum tube.
- ❑ The second generation computers moved from machine language to assembly language.
- ❑ Examples include: **IBM 1401, CDC 3600**, etc.



# Generation of Computers (Cont.)

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## ☐ **Characteristics of Second Generation**

- ☐ Used Transistors
- ☐ Medium sized, faster, and more reliable
- ☐ Magnetic core memory
- ☐ Assembly language programming

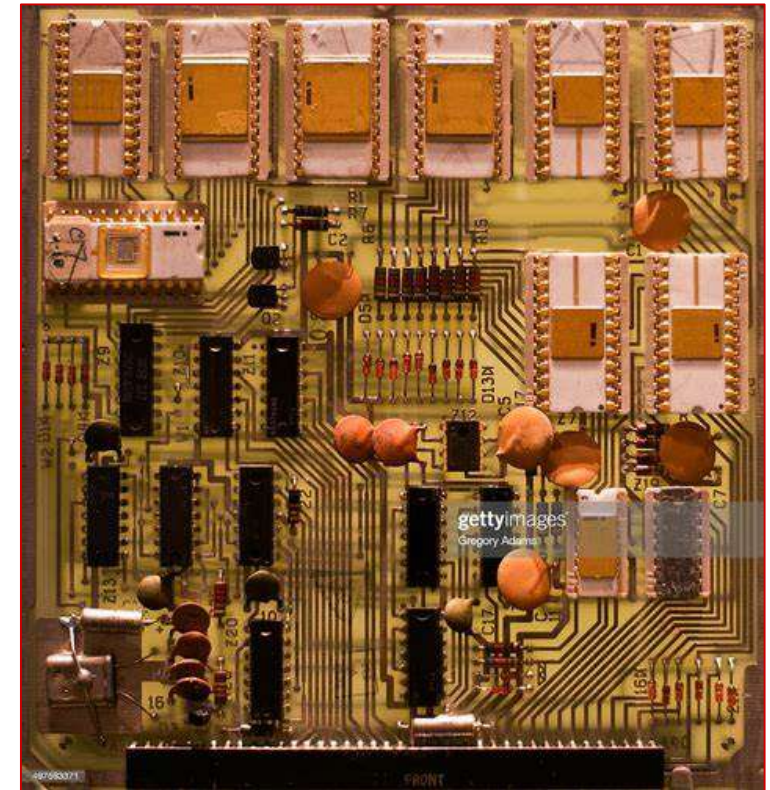


**IBM 1401**

# Generation of Computers (Cont.)

## ❑ The Third Generation: 1964-1971

- ❑ Main Component : Integrated Circuits
- ❑ Integrated Circuits (ICs) replaced transistors in the third generation of computers.
- ❑ ICs were a huge success in advancing computers.
- ❑ The ICs increased the speed and efficiency of the computers.
- ❑ A single IC contained a huge number of transistors. This reduced both the size and cost of computers.
- ❑ Examples include: **IBM System/360, IBM 370**, etc.





# Generation of Computers (Cont.)

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## ☐ **Characteristics of Third Generation**

- ☐ Integrated circuits (ICs) were used
- ☐ Faster and small-sized computers
- ☐ Operating systems were introduced in this generation
- ☐ High-level programming languages were supported by this generation of computers



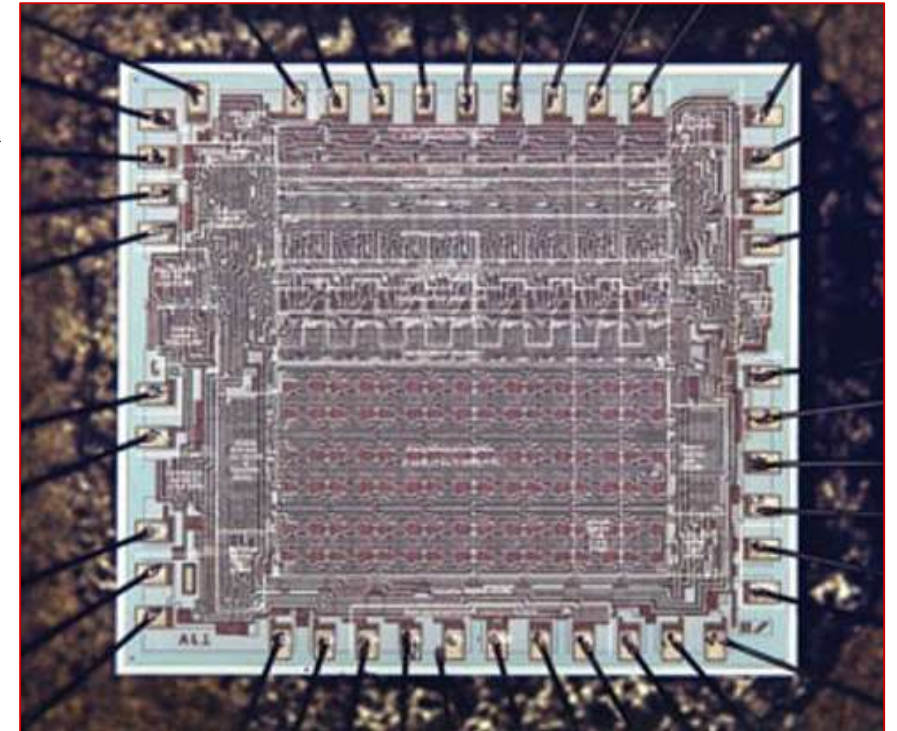
**IBM System/360**

# Generation of Computers (Cont.)

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## ❑ The Fourth Generation: 1971-present

- ❑ Main Component : Microprocessor
- ❑ The fourth generation used microprocessor along with ICs.
- ❑ A single microprocessor contained hundreds/thousands of transistors.
- ❑ It led to the invention of personal computers.
- ❑ Examples include: **DEC 10, PDP 11**, etc.



# Generation of Computers (Cont.)

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## ❑ Characteristics of Fourth Generation

- ❑ Used Microprocessors and Integrated Circuits
- ❑ Personal computers (PCs) were developed in this era
- ❑ Graphical user interfaces (GUI) based operating systems
- ❑ Networking and internet emergence



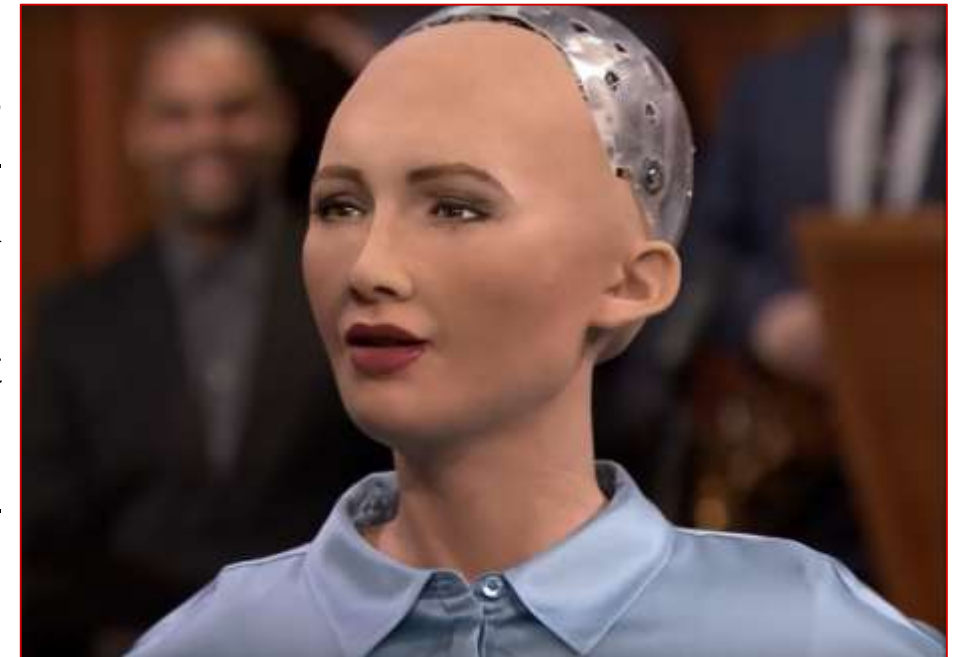
DEC 10

# Generation of Computers (Cont.)

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## ❑ The Fifth Generation: Present and Beyond

- ❑ Main Component : Artificial Intelligence
- ❑ Artificial Intelligence (AI) refers to making computers capable of human-like intelligence and decision-making processes through the use of algorithms and data.
- ❑ Robots and Smart systems are the major development of this generation.
- ❑ These computers are able of learning and self-organizing.
- ❑ Examples include: **AI robots, self driving cars, etc.**



**Sophia: The AI robot**



# Generation of Computers (Cont.)

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## ☐ **Characteristics of Fifth Generation**

- ☐ Artificial Intelligence (AI)
- ☐ Supercomputers
- ☐ Parallel processing
- ☐ Natural language processing



**Self Driving Car**

# Any Question?

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# Thank you

# Information and Communication Technology

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Lecture : 03

Department of Computer Science  
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Fall 2025 Semester  
BS Course: ICT

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  - ❑ Super Computers
  - ❑ Mainframe Computers
  - ❑ Mini Computers
  - ❑ Micro Computers
  - ❑ Embedded Computers

# Classification of Computers (Data Processing)

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- ❑ Computers can be categorized into different types based on the type of data they process.
- ❑ The primary types of computers with respect to data processing are:
  - ❑ Analog Computers
  - ❑ Digital Computers
  - ❑ Hybrid Computers

# Analog Computers

- ❑ Analog Computers are particularly designed to process analog data.
- ❑ Continuous data that changes continuously and cannot have discrete values are called analog data such as electrical, mechanical, hydraulic quantities.
- ❑ Analog computers operate on continuous signals.
- ❑ These computers can directly accept the data from the measuring device.
- ❑ It's output is also analog.



# Digital Computers

- ❑ Digital computers can easily perform calculations and logical operations at high speed.
- ❑ Digital computers process discrete data, which is represented using binary code (0s and 1s).
- ❑ It takes raw data as input, processes it and produce the final output.
- ❑ It only understands the binary input 0 and 1.
- ❑ All modern computers, like laptops, desktops including smartphones are digital computers.



# Hybrid Computers

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- ❑ Hybrid computer is a combination of both analog and digital computers.
- ❑ Hybrid computers are fast like analog computers and have memory and accuracy like digital computers.
- ❑ So, it has the ability to process both continuous and discrete data.
- ❑ Computers used in medical field are hybrid computers i.e. ECG, etc.





# Classification of Computers (Size and Speed)

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- ❑ Computers can be categorized into various types based on their size and processing speed.
- ❑ Here are some common categories:
  - ❑ Super Computers
  - ❑ Mainframe Computers
  - ❑ Mini Computers
  - ❑ Micro Computers
    - ❑ Portable Computers
  - ❑ Embedded Computers

# Super Computers

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- ❑ Supercomputers are the biggest and fastest computers (in terms of speed of processing data).
- ❑ Supercomputers are designed such that they can process a huge amount of data, like processing trillions of instructions in just a second.
- ❑ This is because of the thousands of interconnected processors in supercomputers.
- ❑ It is basically used in scientific and engineering applications such as weather forecasting, scientific simulations, and nuclear energy research.
- ❑ It was first developed by Roger Cray in 1976.



# Super Computers

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- **Size:** These computers are very large and often housed in specialized facilities.
- **Processing Speed:** These are extremely fast, capable of performing trillions of calculations per second (FLOPS - Floating Point Operations Per Second).
- **Purpose:** These are used for complex scientific simulations, weather forecasting, nuclear research, and other computationally intensive tasks.

# Mainframe Computers

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- ❑ Mainframe computers are designed in such a way that they can support hundreds or thousands of users at the same time.
- ❑ It can also run multiple programs simultaneously. So, they can execute different processes simultaneously.
- ❑ All these features make the mainframe computer ideal for big organizations like banking, telecom sectors, etc., which process a high volume of data in general.



# Mainframe Computers

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- **Size:** These are large and typically found in data centers.
- **Processing Speed:** High processing speed, designed for handling large-scale data processing and multitasking for multiple users.
- **Purpose:** Used in large organizations for tasks like banking, financial transaction processing, and telecom.

# Mini Computers

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- ❑ Minicomputer is a medium size multiprocessing computer.
- ❑ This type of computer supports 4 to 200 users at one time.
- ❑ Minicomputers are used in places like institutes or departments for different work like billing, accounting, inventory management, etc.
- ❑ It is smaller than a mainframe computer but larger in comparison to the microcomputer.

# Mini Computers

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- **Size:** Smaller than mainframes but larger than most personal computers.
- **Processing Speed:** Moderate, suitable for medium-range computing.
- **Purpose:** Used in small organizations.

# Micro Computers

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- ❑ A microcomputer is a small, relatively inexpensive computer.
- ❑ These type of computers are usually used for personal use. That's why, it is also called personal computers.
- ❑ Microcomputers are used in places like home, colleges or organizations.
- ❑ It is used for different types of work like personal use, education, clerical work, etc.
- ❑ These include Desktop Computers, Laptops, smartphones, etc.





# Micro Computers

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- **Size:** These are smaller in size and comes in various shapes.
- **Processing Speed:** Processing speed is normal and is suitable for day-to-day tasks.
- **Purpose:** Micro computers fulfill the needs of personal use.

\* A portable computer usually refers to a very light and compact personal computer such as a laptop, subnotebook or handheld PC.



# Embedded Computers

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- ❑ An embedded computer functions as part of a larger device or system, rather than being used as a standalone computer.
- ❑ Typically, an embedded computer performs a specific function.
- ❑ It is used in smart systems like, smartwatches, smart speakers, cars, etc.



# Embedded Computers

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- **Size:** These are tiny and integrated into other devices.
- **Processing Speed:** Varied, according to the need of specific application.
- **Purpose:** Found in everyday devices like appliances, cars, and medical equipment devices to control and monitor various functions.

# Any Question?

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# Thank you

# Information and Communication Technology

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Lecture : 04

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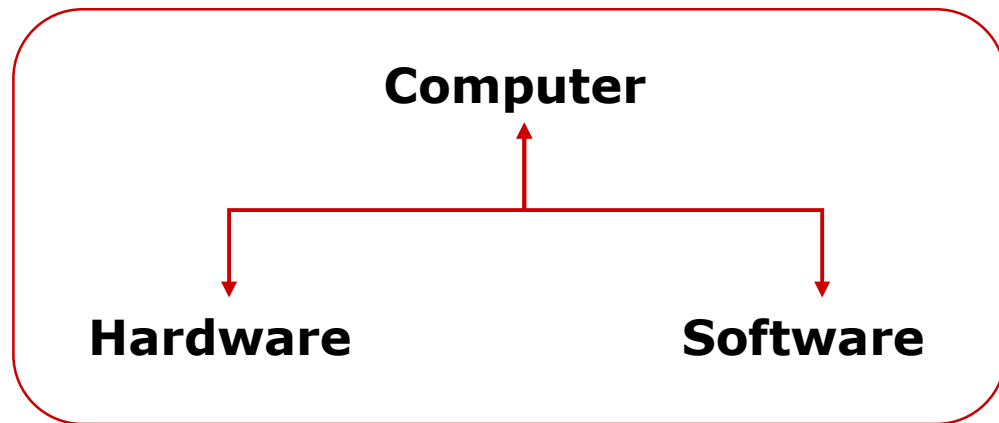
# Contents

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- ❑ Computer System
  - ❑ Computer Hardware
    - ❑ Input Devices
      - ❑ Examples
    - ❑ Output Devices
      - ❑ Examples

# Computer System

- A computer system is a collection of hardware and software components designed to perform various tasks and processes data.



# Hardware

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- ❑ Computer hardware is a physical device/part of computers that we can see and touch.
- ❑ For e.g. Monitor, Central Processing Unit, Mouse, etc.
- ❑ Using these devices, we can control computer operations like input and output.
- ❑ Hardware components are further divided into the following categories, which are:
  - ❑ **Input Devices**
  - ❑ **Output Devices**
  - ❑ **Computer Memory**
    - ❑ **Primary memory and its types**
    - ❑ **Secondary memory and its types**
  - ❑ **Central processing system (CPU)**
  - ❑ **Motherboard**



# Input Devices

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- ❑ Input devices are those devices with the help of which the user interacts with the computer.  
Or,
- ❑ With the help of input devices, the user enters the data or information into the computer.
- ❑ This information or data is accepted by the input devices and converted into a computer-acceptable format.

# Input Devices (Cont.)

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- ❑ **Keyboard:** It is the most common and main input device for computers. The data is inputted by typing on the keyboard. It consists of 104 keys in total. It contains numeric keys, alphabet keys, and different function keys as well. The keyboard layout used today is called QWERTY layout keyboard.
- ❑ **Mouse:** A mouse is a kind of pointing device which controls the cursor on the screen. It has functional keys like left, middle, and right buttons. With mouse you can select an object on the screen.



# Input Devices (Cont.)

- ❑ **Scanner:** As the name suggests, it scans images, documents, etc., and converts them into digital form and that can be further edited and used.
- ❑ **Track Ball:** It is a device much like an upside-down mouse. As the trackball remains stationary and the user moves the ball in various directions, it affects the screen movements directly.



# Input Devices (Cont.)

- ❑ **Light Pen:** It is a light-sensitive device and it is touched to the CRT screen where it can detect, pixels on the screen as it passes by and, with the help of this user can draw anything.
- ❑ **Microphone:** It can be attached to a computer system to record sounds. It converts human speech or voice into electrical signals. This electrical signal is processed by the computer.



# Input Devices (Cont.)

- ❑ **Optical Character Reader:** It is used to detect alphanumeric characters that are written or printed on paper. This light is absorbed by the dark areas and reflected by the light areas. Then this reflected light is received by the photocells. It is like a scanner.
- ❑ **Bar Code Reader:** It is used to read bar codes and convert them into electric pulse which is further processed by the computer. Here, the barcode is data that is coded into white and black lines.



# Output Devices

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- ❑ These are the devices that are used to display the output of any task given to the computer in human-readable form.
- ❑ Through these devices computer can output data or information to the user.

# Output Devices (Cont.)

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- ❑ **Monitor:** The monitor is the main output device. It is also called VDU(visual display unit) and it looks like a TV screen. The Monitor displays the information from the computer. It is used to display text, video, images, etc.
- ❑ **Types:**
- ❑ **Cathode-Ray Tube (CRT):** This type of monitor is based on a cathode ray tube. In which the cathode ray tube generates a beam of electrons with the help of electron guns they strike on the inner surface of phosphorescent of the screen to generate images. The main parts of the CRT monitor are the electron gun, fluorescent screen, glass envelope, deflection plate assembly, and base.



# Output Devices (Cont.)

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- ❑ **LCD Monitor:** LCD is abbreviation of ‘Liquid Crystal Display’ made of liquid crystals.
- ❑ It requires less space, consumes less electricity, and produces relatively less heat than an old CRT monitor.
- ❑ LCD monitors are thinner and much lighter in size and weight than CRT monitors.
- ❑ Due to this, it competes with LED and OLED in the market.





# Output Devices (Cont.)

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- ❑ **LED Monitor:** LED's full form is 'Light Emitting Diode'.
- ❑ It is the latest innovation in today's market competing with LCDs and Plasma Monitors.
- ❑ These types of monitors are slightly curved or flat panel displays that use light-emitting diodes for backlighting on the screen.
- ❑ The advantage of LEDs is that they produce images with higher contrast and vivid colors.
- ❑ In addition, the LEDs are more durable as compared to LCD and CRT Monitors.



# Output Devices (Cont.)

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- ❑ **OLED Monitor:** OLED stands for “Organic Light Emitting Diode“. As the name suggests, it is made of organic material.
- ❑ This is also the latest display technology used in displays of television, computer screen, game consoles, PDAs, or even in the latest smartphones.
- ❑ The OLED display is considered great for watching movies.
- ❑ OLED Monitors are considered the best display technology ever because of their characteristics like **wide viewing angles, picture quality, outstanding contrast levels, fast refresh rate, and perfect contrast and brightness.**



# Output Devices (Cont.)

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- ❑ **Plasma Display:**
- ❑ The Plasma monitor panel is made of Plasma technology. It is another latest type of computer monitor technology.
- ❑ Display of plasma made with cells. These cells are filled with ‘electrically charged Ionized Gas’. Such cells are called Plasma.
- ❑ A plasma display offers a **good refresh rate, superior performance, and a much wide viewing angle** as compared to LCDs.
- ❑ It produces a superior contrast ratio.



# Output Devices (Cont.)

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- ❑ **Speakers:** It is a very common output device and it gives sound as an output. Speaker is generally used to play music or anything having sound.
- ❑ Speakers can be internal or external.



# Output Devices (Cont.)

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- ❑ **Printer:** A printer is an output device that prints data (documents, images, etc.) from the computer on a paper. There are both colored printers and black & white printers. Further, there are two types of printers.
  1. **Impact Printers**
  2. **Non-impact Printers**



# Output Devices (Cont.)

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## 1. Impact Printers

- ❑ In impact printers, characters are printed on the ribbon, which is then smashed on to the paper. Or we can say that such type of printer uses a print head or hammer to print the data on the paper.
- ❑ Some of the types of impact printers are:
  - i. **Dot matrix printer**
  - ii. **Daisy wheel printer**
  - iii. **Line printer**
  - iv. **Chain printer**

# Output Devices (Cont.)

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- ❑ Impact printers have the following characteristics:
  1. Extremely low consumable costs.
  2. Fairly noisy
  3. It's perfect for large-scale printing because of its inexpensive cost.
  4. Physical contact with the paper is required to form an image.

# Output Devices (Cont.)

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## 2. Non-impact Printers

- ❑ Non-impact printers print characters without the use of a ribbon.
- ❑ These printers are often known as page printers because they print a full page at a time.
- ❑ Some of the types of non-impact printers are:
  - i. **Laser printer**
  - ii. **Inkjet printer**



# Output Devices (Cont.)

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- ❑ Non-impact printers have the following characteristics:
  1. Quicker.
  2. They don't produce much noise.
  3. Superior quality.
  4. Supports a wide range of fonts and character sizes.

# Output Devices (Cont.)

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- ❑ **Plotter:** It is similar to a printer but plotters are large in size. A plotter is used to generate large images, drawings, etc. on paper. These are in high-quality and large in size.
- ❑ It is used to print large maps, architectural drawings, large-format printing, and create pictures, 3D postcards, advertising signs, charts, and various designs.



# Output Devices (Cont.)

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- ❑ **Projector**
- ❑ A projector is a device that allows users to project their output onto a large area, such as a screen or a wall. It can be used to project the output of a computer. It magnifies texts, photos, and movies using light and lenses. As a result, it's an excellent output device for giving presentations or teaching big groups of people.



# Any Question?

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# Thank you

# Information and Communication Technology

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Lecture : 05

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# Contents

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- ❑ Computer Memory
  - ❑ Primary Memory
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    - ❑ ROM
    - ❑ Cache
  - ❑ Secondary Memory
    - ❑ Magnetic Tapes
    - ❑ Magnetic Disks
    - ❑ Optical Disks

# Computer Memory

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- ❑ Computer memory refers to the hardware component of a computer system that is used to store data and instructions.
- ❑ Data in memory is accessed by central processing unit (CPU) to process and manipulate.
- ❑ It allows the computer to store and retrieve information for various tasks and processes.
- ❑ Computer memory comes in several types, each with its own characteristics and purposes.
  - ❑ Primary memory
  - ❑ Secondary memory

# Primary memory

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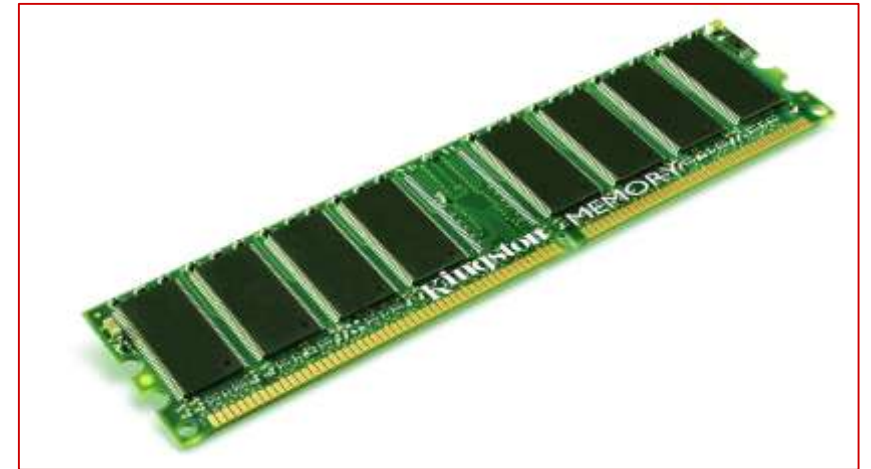
- ❑ It is also known as the main memory of the computer system.
- ❑ It is used to store data and programs/instructions during computer operations.
- ❑ It uses semiconductor technology and hence is commonly called semiconductor memory.
- ❑ Primary memory is of three types:
  - ❑ RAM (Random Access Memory)
  - ❑ ROM (Read Only Memory)
  - ❑ Cache Memory



# RAM (Random Access Memory)

---

- ❑ RAM (Random Access Memory) is the memory used by the CPU for storing data and programs temporarily.
- ❑ It is a read/write memory which stores data until the computer running. As soon as the machine is switched off, data in RAM is erased.
- ❑ Also called volatile memory.
- ❑ RAM is small in size.
- ❑ The data it can hold is up to few giga bytes.
- ❑ RAM is categorized into two types -
  - ❑ Static RAM (SRAM)
  - ❑ Dynamic RAM (DRAM)



# RAM (Random Access Memory)

---

- ❑ **SRAM (Static RAM):**

- ❑ This type of RAM is faster and more reliable than DRAM.
- ❑ It doesn't need to be refreshed like DRAM, but it is more expensive to produce.
- ❑ It is typically used in smaller amounts for **cache memory**.

- ❑ **DRAM (Dynamic RAM):**

- ❑ This is the most common type of RAM used in computers.
- ❑ It needs to be refreshed thousands of times per second, as it stores each bit of data in a separate capacitor within an integrated circuit.

# RAM (Random Access Memory)

---

- ❑ **SDRAM (Synchronous Dynamic RAM):**
- ❑ SDRAM is synchronized with the computer's system bus, allowing for faster data transfer rates.
- ❑ It has been widely used in the past but has largely been succeeded by newer types like DDR.
  
- ❑ **DDR SDRAM (Double Data Rate Synchronous Dynamic RAM):**
- ❑ DDR SDRAM transfers data at double rate compared to SDRAM. There are several generations of DDR, such as DDR2, DDR3, and DDR4, each offering improvements in speed and efficiency.

# RAM (Random Access Memory)

---

- ❑ **LPDDR (Low Power DDR):** LPDDR is a type of DDR SDRAM designed for lower power consumption, making it suitable for use in mobile devices like smartphones and tablets.
- ❑ **GDDR (Graphics Double Data Rate):** GDDR is a type of specialized SDRAM used in graphics cards. It is optimized for high-bandwidth data transfer and is specifically designed to meet the needs of graphics processing units (GPUs).

# ROM (Read Only Memory)

---

- ❑ From ROM we can only read but cannot write on it.
- ❑ This type of memory is non-volatile.
- ❑ The information is stored permanently in such memories during manufacturing.
- ❑ A ROM stores such instructions that are required to start a computer.
- ❑ ROM chips are not only used in the computer but also in other electronic items like washing machine and microwave oven.



# ROM (Read Only Memory)

---

- ❑ ROM has three types:
  - ❑ PROM (Programmable Read Only Memory)
  - ❑ EPROM (Erasable Programmable Read Only Memory)
  - ❑ EEPROM (Electrically Erasable Programmable Read Only Memory)
  - ❑ Flash Memory

# ROM (Read Only Memory)

---

- ❑ **PROM (Programmable Read-Only Memory):**
- ❑ PROM allows users to write or program data onto the chip after it has been manufactured.
- ❑ However, this process is irreversible, and once programmed, the data cannot be modified.
- ❑ PROM is programmed using a special device called a PROM programmer.
- ❑ **EPROM (Erasable Programmable Read-Only Memory):**
- ❑ EPROM allows for data to be erased and reprogrammed multiple times.
- ❑ Erasure is typically achieved by exposing the memory chip to ultraviolet (UV) light for a specified amount of time.
- ❑ EPROMs are commonly used in situations where occasional updates to the stored data are required.

# ROM (Read Only Memory)

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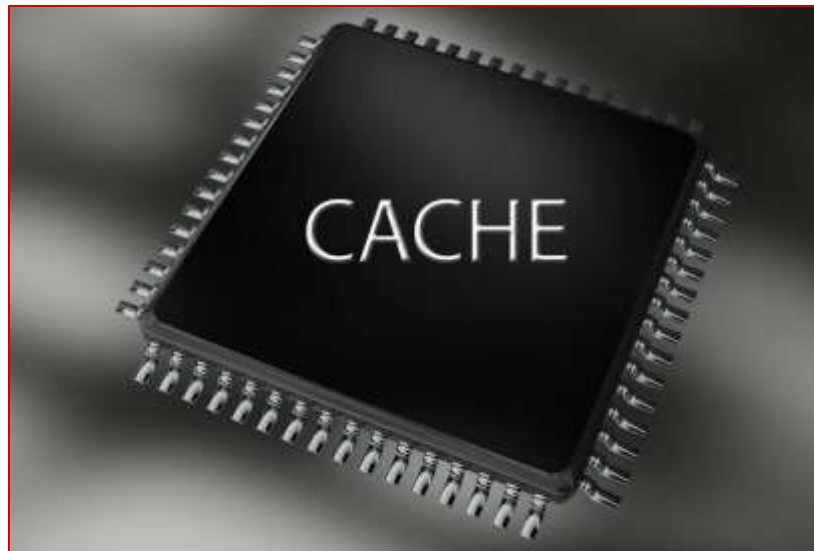
- ❑ **EEPROM (Electrically Erasable Programmable Read-Only Memory):**
- ❑ EEPROM, also known as Flash EEPROM, allows for the erasure and reprogramming of data electrically, without the need for UV light.
- ❑ This makes EEPROM more convenient for updating firmware or other data without removing the chip from the device.
  
- ❑ **Flash Memory:**
- ❑ Flash memory is a type of EEPROM that can be electrically erased and reprogrammed in blocks, rather than one byte at a time.
- ❑ It is commonly used in USB drives, memory cards, solid-state drives (SSDs), and other storage devices.



# Cache Memory

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- ❑ It is a type of high-speed semiconductor memory that can help the CPU run faster.
- ❑ It is placed between the CPU and the main memory, and serves as a buffer.
- ❑ It is used to store the data and programs that the CPU uses most frequently.



# Cache Memory

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- ❑ **L1 Cache (Level 1 Cache):**

- ❑ L1 cache is the smallest and fastest cache directly integrated into the CPU core.
- ❑ It is split into two parts: instruction cache (L1i) for storing instructions and data cache (L1d) for storing data.
- ❑ L1 cache provides extremely fast access to frequently used instructions and data.

- ❑ **L2 Cache (Level 2 Cache):**

- ❑ L2 cache is larger than L1 cache and is located outside the CPU core but still on the same chip.
- ❑ It serves as a secondary cache to support the L1 cache.
- ❑ L2 cache is slower than L1.

# Cache Memory

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- ❑ **L3 Cache (Level 3 Cache):**
- ❑ L3 cache is larger than L2 cache and is shared among the processor cores in a multi-core system.
- ❑ It provides a common cache for multiple cores to access frequently used data.
- ❑ L3 cache is slower than L1 and L2 cache.

# Secondary Memory

---

- ❑ It is also known as auxiliary memory and backup memory.
- ❑ It is a non-volatile memory and used to store a large amount of data or information.
- ❑ The data or information stored in secondary memory is permanent.
- ❑ It is slower than primary memory.
- ❑ A CPU cannot access secondary memory directly.
- ❑ The data/information from the auxiliary memory is first transferred to the main memory, and then the CPU can access it.

# Secondary Memory

---

- ❑ **Magnetic Disks:** A magnetic disk is a circular metal plate which are coated with magnetic material. The disc is used on both sides. Bits are stored on the magnetized surfaces.
- ❑ Example: Floppy disk drive, Hard disk drive.

# Secondary Memory

---

- ❑ **Floppy Disk:** A floppy disk consists of a magnetic disc in a square plastic case. It was used to store data and to transfer data from one device to another device. Floppy disks were available in two sizes (a) **Size: 3.5 inches**, (b) **Size: 5.25 inches**. To use a floppy disk, our computer needs to have a floppy disk drive. This storage device becomes obsolete now and has been replaced by CDs, DVDs, and flash drives.



# Secondary Memory

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- ❑ **Hard Disk:** A hard disk is a part of a unit called a hard disk drive. It is used to storing a large amount of data. Hard disks or hard disk drives come in different storage capacities.(like 256 GB, 500 GB, 1 TB, and 2 TB, etc.). It is created using the collection of discs known as platters. The platters are placed one below the other. They are coated with magnetic material. Each platter consists of a number of invisible circles and each circle having the same center called tracks. Hard disk is of two types:
  - ❑ **Internal hard disk**
  - ❑ **External hard disk.**



# Secondary Memory

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- ❑ **Solid-state disk:** It is also known as SSD. It is a storage device that is used to store and access data. It is faster, does noiseless operations(because it does not contain any moving parts like the hard disk), consumes less power, etc. It is a great replacement for standard hard drives in computers and laptops.





# Secondary Memory

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- ❑ **Optical Disks:** It is a laser-based storage medium that can be used to store data. It is reasonably priced and has a long lifespan. The optical disc is read by CD Drive or DVD Drive.
- ❑ Example: Compact Disk (CD), Digital Versatile Disk (DVD), Blu-ray Disks

# Secondary Memory

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- ❑ **Compact Disc:** A Compact Disc (CD) is a commonly used secondary storage device. It contains tracks and sectors on its surface. Its shape is circular and is made up of polycarbonate plastic. The storage capacity of CD is up to 700 MB of data. For a CD, we require a CD-ROM to read the data present on it. CD is of three types:
  - ❑ **CD-ROM (Compact Disc Read-Only Memory):**
  - ❑ **CD-R (compact disc recordable):**
  - ❑ **CD-RW (compact disc rewritable):**



# Secondary Memory

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- ❑ **Digital Versatile Disc:** A Digital Versatile Disc also known as DVD it looks just like a CD, but the storage capacity is greater compared to CD, it stores up to 4.7 GB of data. DVD-ROM drive is needed to use DVD on a computer. The video files, like movies or video recordings, etc., are generally stored on DVD. DVD is of three types:
  - ❑ **DVD-ROM(Digital Versatile Disc Read-only):**
  - ❑ **DVD-R(Digital Versatile Disc Recordable):**
  - ❑ **DVD-RW(Digital Versatile Disc Rewritable and Erasable):**



# Secondary Memory

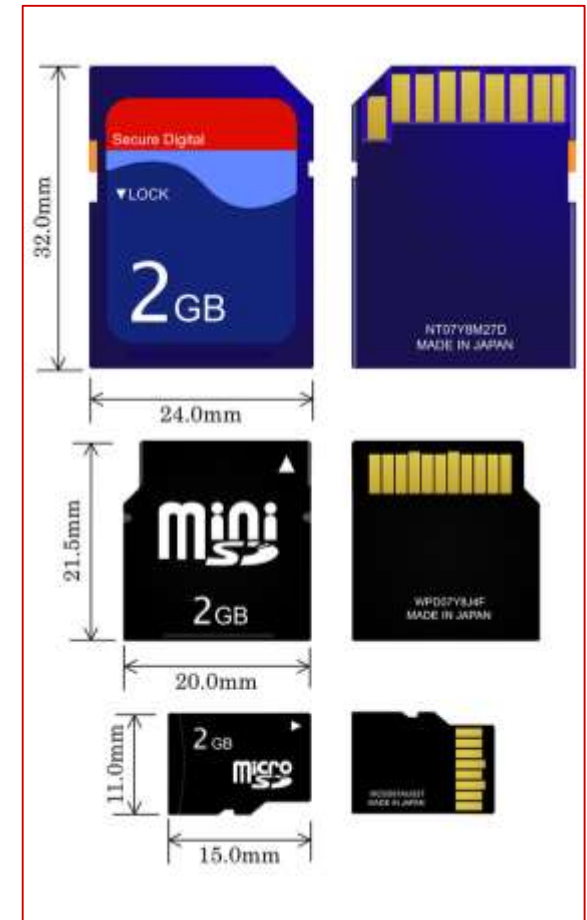
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- ❑ **Blu-ray Disc:** A Blu-ray disc looks just like a CD or a DVD but it can store up to 25 GB data. If you want to use a Blu-ray disc, you need a Blu-ray reader. The name Blu-ray is derived from the technology that is used to read the disc ‘Blu’ from the blue-violet laser and ‘ray’ from an optical ray.



# Secondary Memory

- ❑ **SD Card:** It is known as a Secure Digital Card. It is generally used in portable devices like mobile phones, cameras, etc., to store data. It is available in different sizes like 1 GB, 2 GB, 4 GB, 8 GB, 16 GB, 32 GB, 64 GB, etc. To view the data stored in the SD card you can remove them from the device and insert them into a computer with help of a card reader. The data stores in the SD card is stored in memory chips (present in the SD Card) and it does not contain any moving parts like the hard disk.



# Secondary Memory

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- ❑ **Flash Drive:** A flash drive or pen drive comes in various storage capacities, such as 1 GB, 2 GB, 4 GB, 8 GB, 16 GB, 32 GB, 64 GB, up to 1 TB. A flash drive is used to transfer and store data. To use a flash drive, we need to plug it into a USB port on a computer. As a flash drive is easy to use and compact in size, Nowadays it is very popular.



# Any Question?

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# Thank you

# Information and Communication Technology

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Lecture : 06

Department of Computer Science  
GPGC SWABI (Khyber Pakhtunkhwa)

Fall 2025 Semester  
BS Course: ICT

Course Instructor :  
Mr. Ihtisham Ul Haq  
Lecturer (CS)  
GPGC Swabi



# Contents

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- ❑ Central processing system (CPU)
  - ❑ Arithmetic Logic Unit (ALU)
  - ❑ Control Unit (CU)
  - ❑ Registers
- ❑ Motherboard

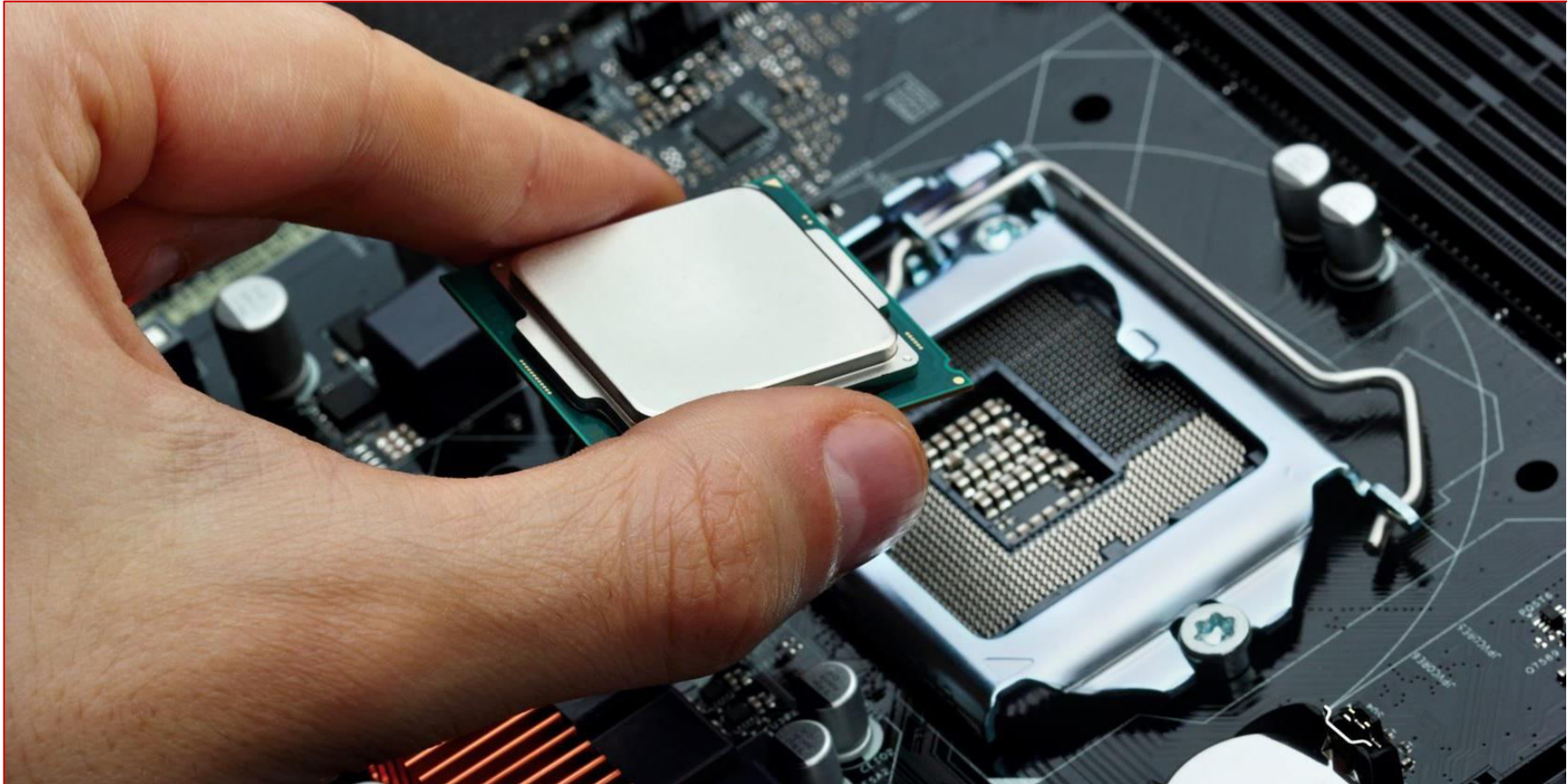
# Central processing system (CPU)

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- ❑ A Central Processing Unit is the most important component of a computer system.
- ❑ It is called the brain of the computer.
- ❑ All types of data processing operations and all the important functions of a computer are performed by the CPU.
- ❑ It helps input and output devices to communicate with each other and perform their respective operations.
- ❑ CPU can perform various data processing operations.

# Central processing system (CPU)

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# Central processing system (CPU)

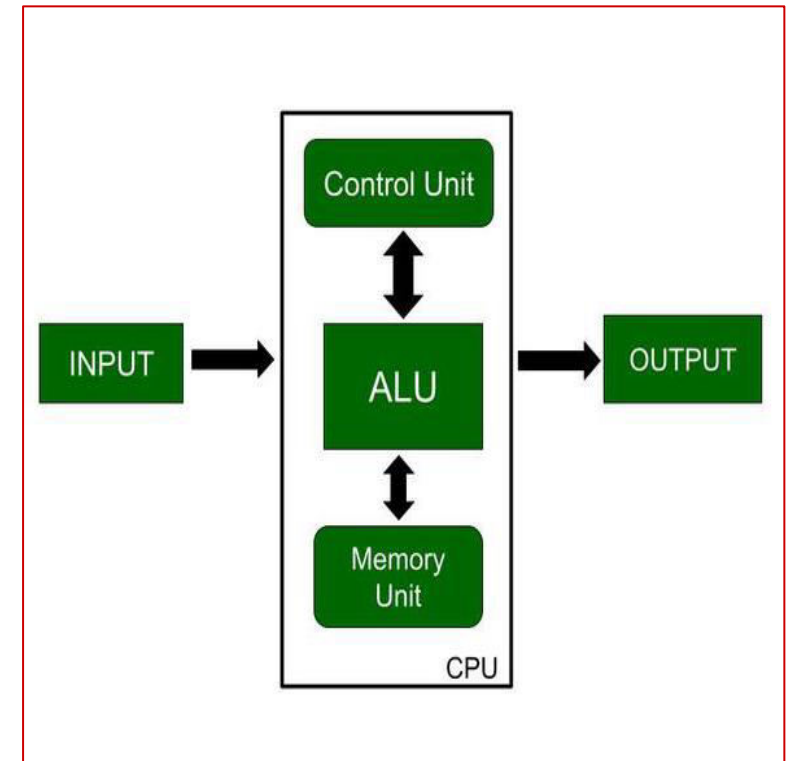
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- ❑ CPU can store data, instructions, programs, and intermediate results.
- ❑ CPU consists of three parts:
  - Arithmetic Logic Unit (ALU)
  - Control Unit (CU)
  - Registers



# Arithmetic Logic Unit (ALU)

- ❑ ALU (Arithmetic Logic Unit) is responsible for performing arithmetic and logical operations.
- ❑ By arithmetic operations, we mean operations like addition, subtraction, multiplication, and division, and all these operation are performed by ALU.
- ❑ By Logical operations, we mean operations like selecting, comparing, matching, and merging the data, and all these are performed by ALU.



# Control Unit (CU)

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- ❑ Control unit controls the operations of all parts of the computer.
- ❑ It does not carry out any data processing operations.
- ❑ It takes instructions from the memory unit and then decodes the instructions, after that it executes those instructions.
- ❑ So, it controls the functioning of the computer.
- ❑ It's main task is to maintain the flow of information across the processor.

# Registers

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- ❑ Registers are high-speed storage locations within a CPU.
- ❑ They are used for temporary data storage and processing during program execution.
- ❑ Registers play a vital role in arithmetic and logical operations performed by the CPU.
- ❑ Common types of registers include **data registers, address registers, and control registers.**
- ❑ Registers enable faster data access and manipulation compared to accessing main memory.

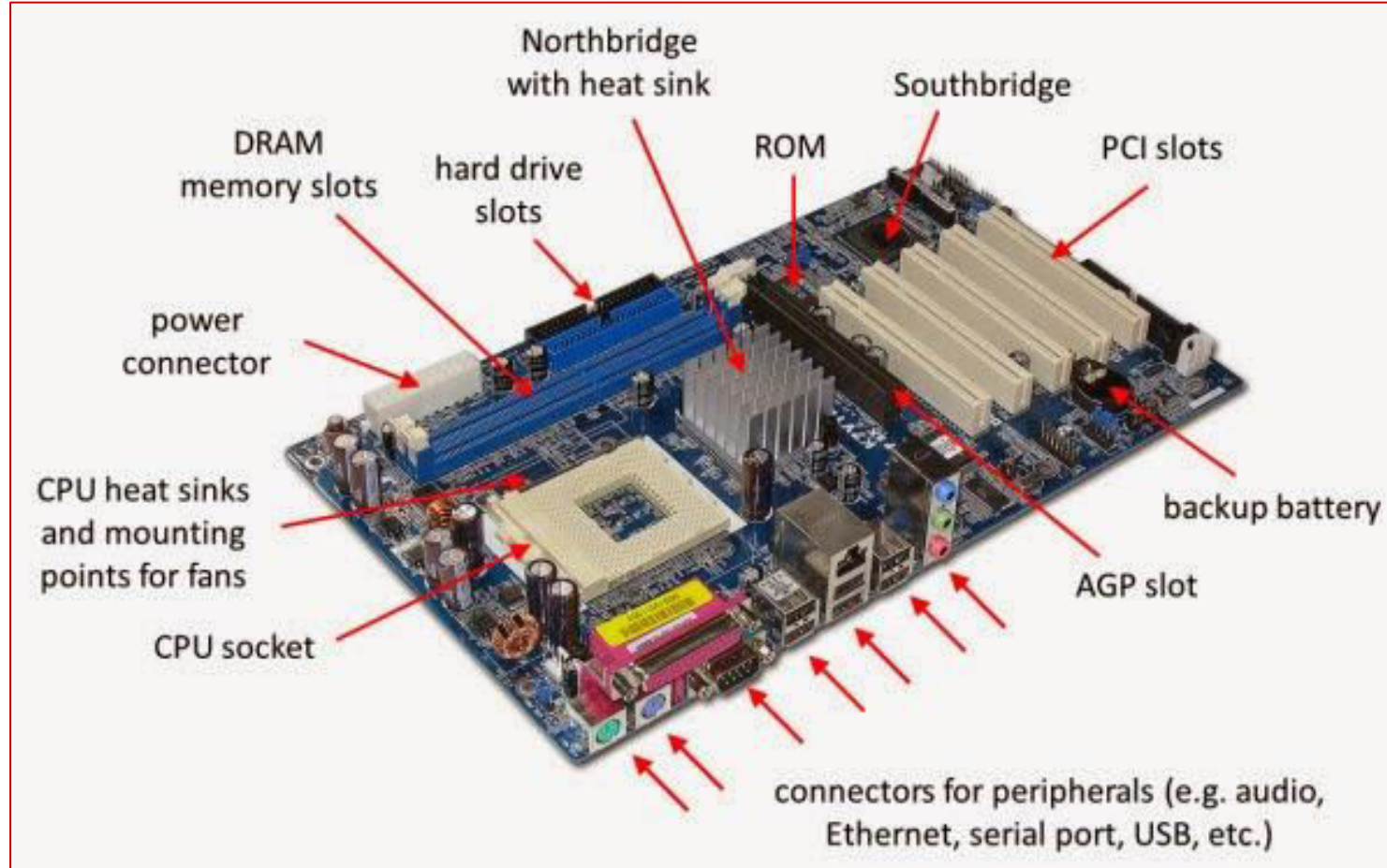
# Motherboard

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- ❑ A motherboard is a critical component of a computer system that serves as the main circuit board.
- ❑ It is a central hub for connecting various hardware components.
- ❑ It gets power from the power supply component and then distributes it.
- ❑ It connects and provides power distribution among the CPU (Central Processing Unit), RAM (Random Access Memory), storage devices, graphics cards, and other essential components.
- ❑ All the input and output devices are also connected to the motherboard.
- ❑ The CPU slot (where CPU is inserted) is located on the motherboard.



# Motherboard



# Motherboard

---

- ❑ **Serial port(COM Port):**
- ❑ A serial port is also called a communication port and they are used for connection of external devices like a modem, mouse, or keyboard (basically in older PCs). There are two versions of it, which are 9 pin model and 25 pin model.
- ❑ **Parallel Port (LPT ports):**
- ❑ Parallel ports are generally used for connecting scanners and printers. It can send several bits at the same time as it uses parallel communication. It is a 25 pin model. It is also known as Printer Port or Line Printer Port.
- ❑ **PS/2 Port:**
- ❑ PS/2 ports are special ports used for connecting old computer keyboard and mouse. It was invented by IBM. In old computers, there are minimum of two PS/2 Ports, each for the keyboard and the mouse. It is a 6 pin mini Din connector.

# Motherboard

---

- ❑ **USB (Universal Serial Bus):**
- ❑ In 1997 USB was first introduced. This can connect all kinds of external USB devices, like external hard disk, printer, scanner, mouse, keyboard, etc.
- ❑ There are minimum of two USB Ports provided in most of the computer systems.
- ❑ These USB Ports are much smarter and more versatile. It also provides plug & plays communication.
- ❑ **VGA Port:**
- ❑ VGA ports also known as Video Graphic Array connector are those which connect the monitor to a computer's video card. VGA port has 15 holes and it is similar to the serial port connector. But VGA Ports have holes in it and the serial port connector has pins in it.

# Motherboard

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- ❑ **Sockets:**
- ❑ Microphones and speakers are connected with the help of Sockets to the sound card of the computer.
- ❑ **FireWire Port:**
- ❑ The FireWire port was developed by Apple. It can transfer large amount of data at very high speed. It is used to connect camcorders and video equipment to the computer. It comes up with three variants which are 4-Pin FireWire, 6-Pin FireWire, and 9-Pin FireWire.

# Motherboard

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- ❑ **Digital Video Interface(DVI) Port:**
- ❑ DVI Port is used to connects LCD(flat panel) monitor to the computer's high-end video graphic cards and it is very popular among video card manufacturers.
- ❑ **Ethernet Port:**
- ❑ Ethernet Port helps to connect to a network and high-speed Internet(provided by LAN or other sources). It connects the network cable to a computer and resides in a Ethernet card. It provides a data travel speed of 10 Mb to 1000 Mb(megabits) per second.

# Any Question?

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# Thank you

# Information and Communication Technology

---

Lecture : 07

Department of Computer Science  
GPGC SWABI (Khyber Pakhtunkhwa)

Fall 2025 Semester  
BS Course: ICT

Course Instructor :  
Mr. Ihtisham Ul Haq  
Lecturer (CS)  
GPGC Swabi

# Contents

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- ❑ Computer Software
  - ❑ System software
    - ❑ Operating System
    - ❑ Device Drivers
    - ❑ Language Processors
  - ❑ Application software
    - ❑ General Purpose Software
    - ❑ Customize Software
    - ❑ Utility Software



# Software

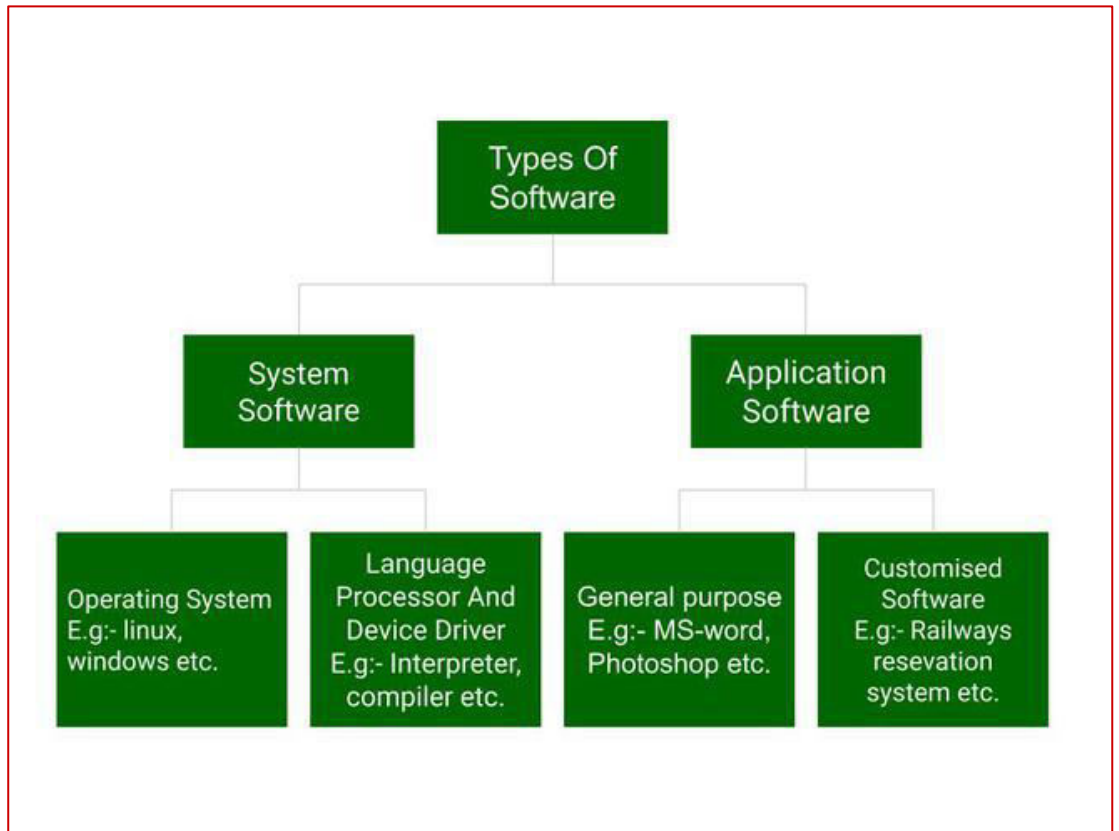
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- ❑ In a computer system, the software is basically a set of instructions or commands that tell a computer what to do.
- ❑ The software is a computer program that provides a set of instructions to execute a user's commands and tell the computer what to do.
- ❑ For example like Microsoft Windows, MS Word, MS PowerPoint, etc.

# Software

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- ❑ Computer software has two categories:
  - System Software
  - Application Software



# System Software

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- ❑ System software is software that directly operates the computer hardware.
- ❑ System software basically controls a computer's internal functioning and also controls hardware devices such as monitors, printers, and storage devices, etc.
- ❑ It is like an interface between hardware and applications software.
- ❑ System software converts the human-readable language into machine language and vice versa.
- ❑ Types:
  - ❑ **Operating Systems**
  - ❑ **Language Processors**
  - ❑ **Device Drivers**

# System Software

---

- ❑ **Operating Systems:** Software that manages computer hardware resources, software resources, and provides user interfaces for efficient interaction with the computer.
- ❑ E.g. MS Windows, Mac OS, Linux, etc.



# System Software

---

- ❑ **Language Processors:** These are programs like compilers and interpreters that translate high-level programming code into machine code to execute it directly.
- ❑ E.g. Visual Studio Code, C Compiler, etc.
- ❑ Language Processors are divided into three types:
  - 1) **Compilers**
  - 2) **Interpreters**
  - 3) **Assembler**

# System Software

---

- ❑ **Device Drivers:** Software that enables communication between the operating system and specific hardware devices, allowing them to function correctly with the computer system.
- ❑ E.g. Audio drivers, Video drivers, etc.
- ❑ **Plug-n-Play?**



# Application Software

---

- ❑ Software that performs specific functions are known as application software.
- ❑ In other words, application software is designed to perform a specific task for end-users.
- ❑ It is a product or a program that is designed only to fulfill users requirements.
- ❑ It includes word processors, spreadsheets, database management, inventory management systems, payroll programs, etc.
- ❑ Types:
  - ❑ **General Purpose Software**
  - ❑ **Customized Software**
  - ❑ **Utility Software**

# System Software

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- ❑ **General Purpose Software:** Software designed to perform a wide range of common computing tasks and functions, suitable for a broad user base.
- ❑ E.g. MS Office, Google Chrome. etc.





# System Software

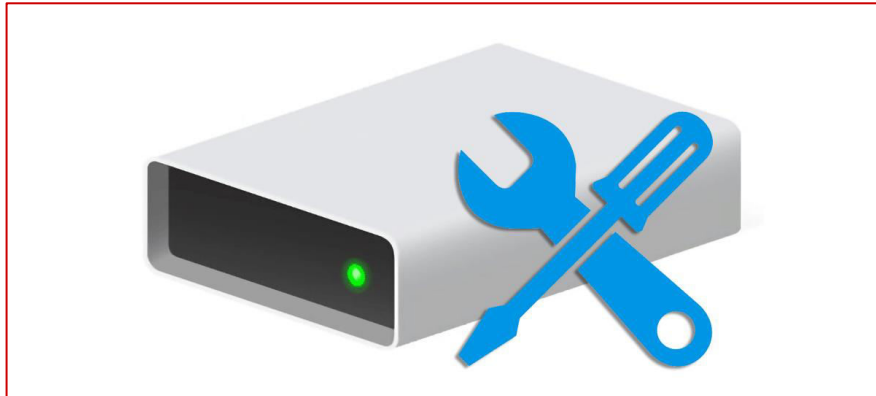
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- ❑ **Customized Software:** Software that fulfills specific needs or requirements of an individual or organization, often developed uniquely for a particular purpose.
- ❑ E.g. student management system, point of sale software, etc.

# System Software

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- ❑ **Utility Software:** Software designed to provide essential system tools and functions.
- ❑ Such as disk management, antivirus, and system optimization etc.



# Any Question?

---

# Thank you

# Information and Communication Technology

---

Lecture : 08

Department of Computer Science  
GPGC SWABI (Khyber Pakhtunkhwa)

Fall 2025 Semester  
BS Course: ICT

Course Instructor :  
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# Contents

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- ❑ Data Communication
  - ❑ Components of Data Communication
  - ❑ Modes of Data Communication
- ❑ Computer Networks
  - ❑ Network Types
- ❑ Network Topology
  - ❑ Network topologies types

# Data Communication

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- ❑ The process of transmitting digital or analog data between devices or systems.
- ❑ Communication is a process or act in which we can send or receive data.
- ❑ Effective data communication requires the use of appropriate protocols, and hardware devices.

# Components of Data Communication

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- ❑ There are five components of data communication.
- 1) **Sender:** The device or system that initiates the data transmission.
- 2) **Receiver:** The device or system that receives the transmitted data.
- 3) **Message:** A message is a piece of information that is to be transmitted from one person to another. It could be a text file, an audio file, a video file, etc.
- 4) **Medium/Channel:** The pathway through which data is transmitted, such as cables or wireless connections.
- 5) **Protocol:** A set of rules and conventions that govern the transmission and reception of data.

# Modes/Types of Data Communication

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❑ The data communication is divided into three different types/modes:

1) **Simplex Communication:** It is one-way communication or we can say that unidirectional communication. In this mode, one device only receives and another device only sends data. Devices use their entire capacity in transmission. For example, TV, radio, Keyboard to CPU, etc.



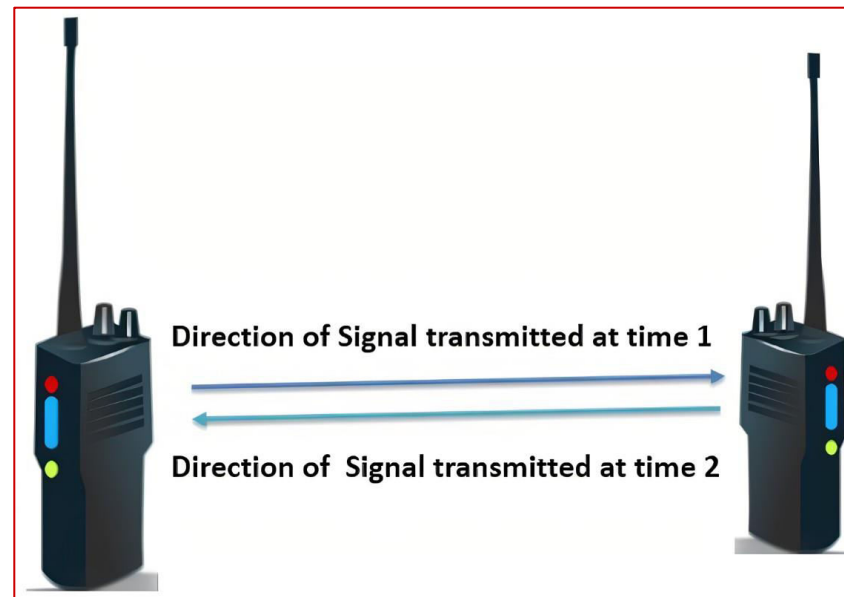
Fig: Simplex Mode of Transmission



# Modes/Types of Data Communication

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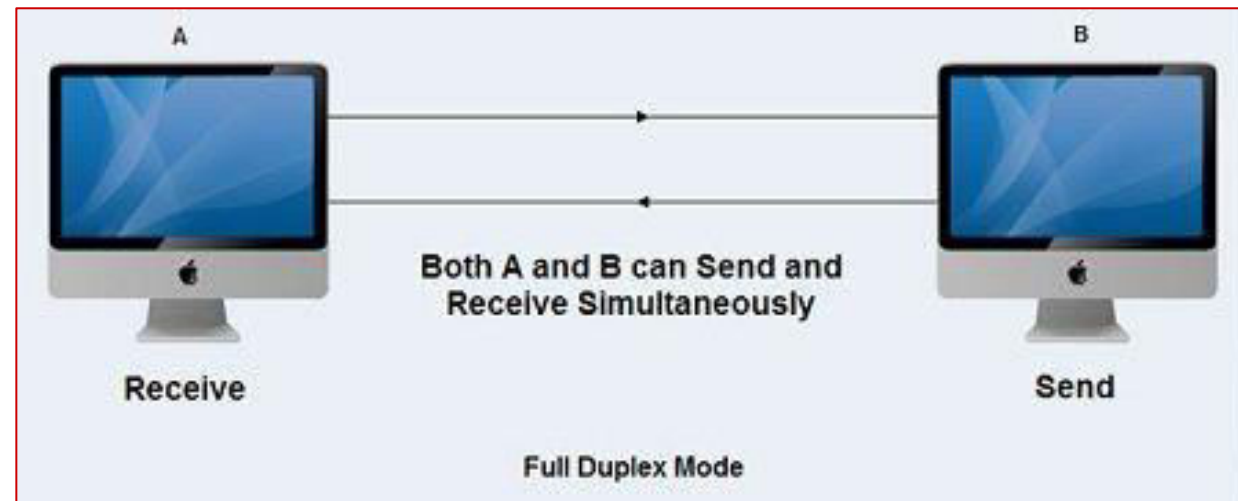
- 2) Half Duplex communication:** It is a two-way communication or we can say that it is a bidirectional communication. In this mode, both the devices can send and receive data but not at the same time. When one device is sending data then another device is only receiving and vice-versa. For example, walkie-talkie.



# Modes/Types of Data Communication

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- 3) **Full-duplex communication:** It is a two-way communication or we can say that it is a bidirectional communication. In this mode, both the devices can send and receive data at the same time. For example, mobile phones, landlines, etc.



# Computer Networks

---

- ❑ A network is a collection of computers and devices that are connected together to enable communication and data exchange.
- ❑ It helps users to communicate more easily.
- ❑ Network Types:
  - ❑ Personal Area Network (PAN)
  - ❑ Local Area Network (LAN)
  - ❑ Metropolitan Area Network (MAN)
  - ❑ Wide Area Network (WAN)

# Personal Area Network (PAN)

---

- ❑ A Personal Area Network (PAN) is a small, short-range network.
- ❑ It is designed for connecting personal devices and peripherals within the immediate small area.
- ❑ It has typically a range of a few meters (around 30 feet or less).
- ❑ PANs are commonly used for connecting devices like smartphones, laptops, tablets, Bluetooth headphones, and other gadgets.



# Local Area Network (LAN)

---

- ❑ A Local Area Network (LAN) is a network of interconnected computers and devices within a limited geographic area, such as a home, office, school, or a small campus.
- ❑ LANs are designed for local and relatively high-speed data communication, allowing connected devices to share resources and information efficiently.
- ❑ LANs provide fast data transfer rates, often reaching gigabit speeds or higher.

# Metropolitan Area Network (MAN)

---

- ❑ A Metropolitan Area Network is a type of computer network that spans a city-sized area.
- ❑ It is larger in geographic scope compared to a Local Area Network (LAN) but smaller than a Wide Area Network (WAN).
- ❑ MANs are designed to interconnect multiple LANs within a metropolitan area.
- ❑ MANs provide relatively high data transfer rates.
- ❑ Typically used by organizations, universities.

# Wide Area Network (WAN)

---

- ❑ A Wide Area Network (WAN) is a type of computer network that spans a large geographic area, often a country, continent, or even the entire world.
- ❑ WAN's are larger than Local Area Networks (LANs) and Metropolitan Area Networks (MANs).
- ❑ WAN's connect many LAN's and WAN's over longer distances.

# Network Topology

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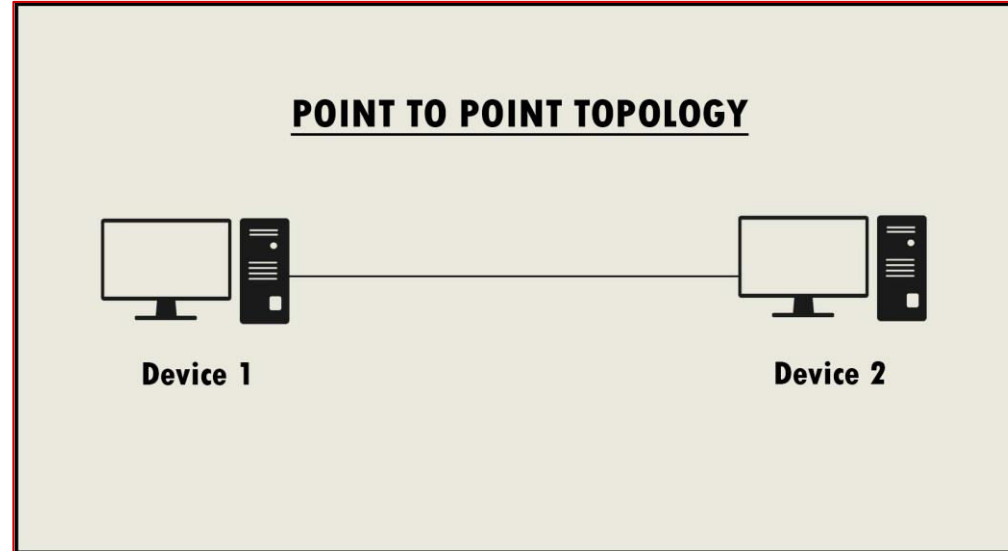
- ❑ Network topology refers to the layout or arrangement of the elements (such as computers, servers, routers, switches, and cables) in a computer network.
- ❑ It defines how these components are connected to one another and how data and information are transmitted within the network.
- ❑ Network topology types are:
  - ❑ **Point-to-Point**
  - ❑ **Bus**
  - ❑ **Ring**
  - ❑ **Star**
  - ❑ **Mesh**



# Network Topology Types

---

- ❑ **Point-to-Point Topology:**
- ❑ Point-to-Point Topology is a type of topology that provides a dedicated connection between two devices only. It is the simplest communication between two nodes, in which one is the sender and the other one is the receiver. Point-to-Point provides high bandwidth.



# Network Topology Types

---

## ☐ **Advantages:**

- ☐ Simple Setup: It's easy to connect just two devices directly.
- ☐ Fast Communication: Data goes straight between devices, so it's quick.
- ☐ No Sharing: The connection is dedicated to only two devices, so no one else can use it.

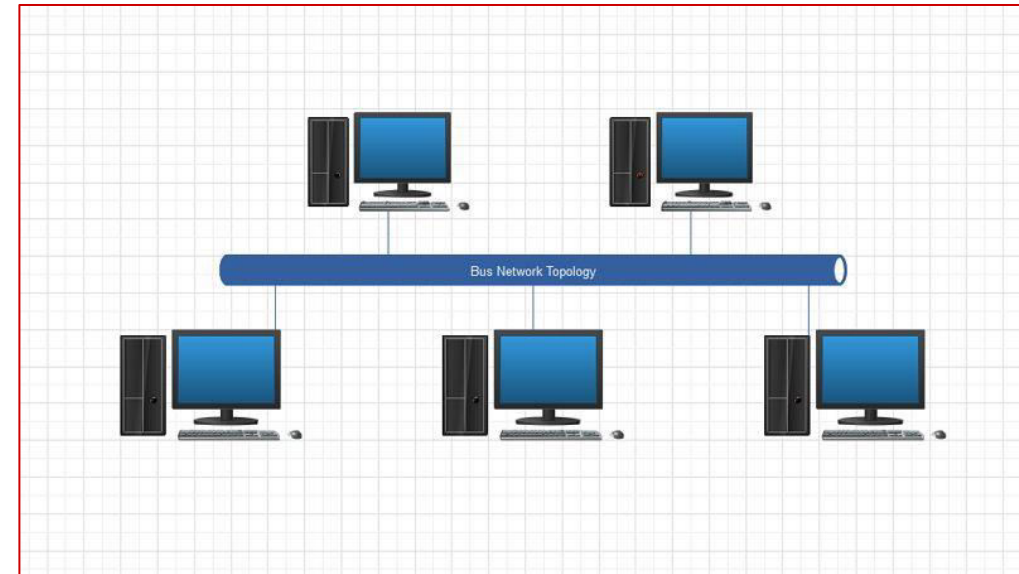
## ☐ **Disadvantages:**

- ☐ Limited Devices: You can't add more devices to this kind of connection.
- ☐ No Backup: If the link between the devices breaks, there's no backup.
- ☐ Less Useful: It is less useful than other types of networks.

# Network Topology Types

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- ❑ **Bus Topology:**
- ❑ It is a network topology in which every computer and network device is connected to a single cable.
- ❑ It is bi-directional and has multipoint connections.
- ❑ It is a non-robust topology because if the backbone fails the topology crashes.
- ❑ It uses one central RJ-45 network cable or coaxial cable.



# Network Topology Types

---

## ☐ **Advantages:**

- ☐ **Simplicity:** Bus topology is easy to install and set up because it uses a single main cable.
- ☐ **Cost-Effective:** It requires less cabling, making it cost-effective for small networks.
- ☐ **Scalability:** It's relatively easy to add new devices to the network by connecting them to the main cable.

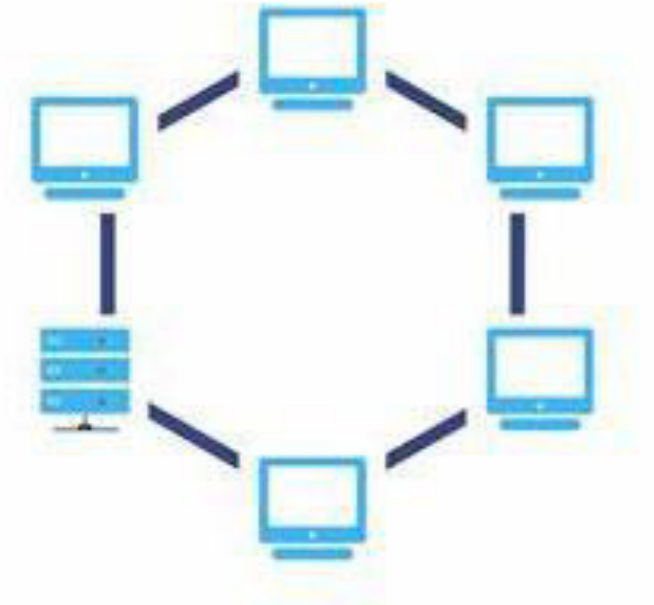
## ☐ **Disadvantages:**

- ☐ **Single Point of Failure:** If the main cable fails, the entire network can go down.
- ☐ **Limited Length and Devices:** There are practical limitations to the length of the main cable and the number of devices you can connect.
- ☐ **Data Collisions:** In a busy network, data collisions can occur, leading to data loss and network inefficiency.

# Network Topology Types

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- ❑ **Ring Topology:**
- ❑ In a ring topology, devices are connected in a circular fashion, where each device is connected to exactly two other devices.
- ❑ Data travels in a unidirectional or bidirectional manner around the ring.
- ❑ Ring topologies are relatively fault-tolerant, but a failure at any point in the ring can disrupt the entire network.



# Network Topology Types

---

## ☐ **Advantage:**

- ☐ **Reliability:** It's usually dependable because data can go in both directions, so if one device fails, it can still work.
- ☐ **Predictable Data Path:** Data always goes in a set order, making it good for things that need precise timing.
- ☐ **Fair Sharing:** Every device gets a fair chance to send data since it follows a fixed order.

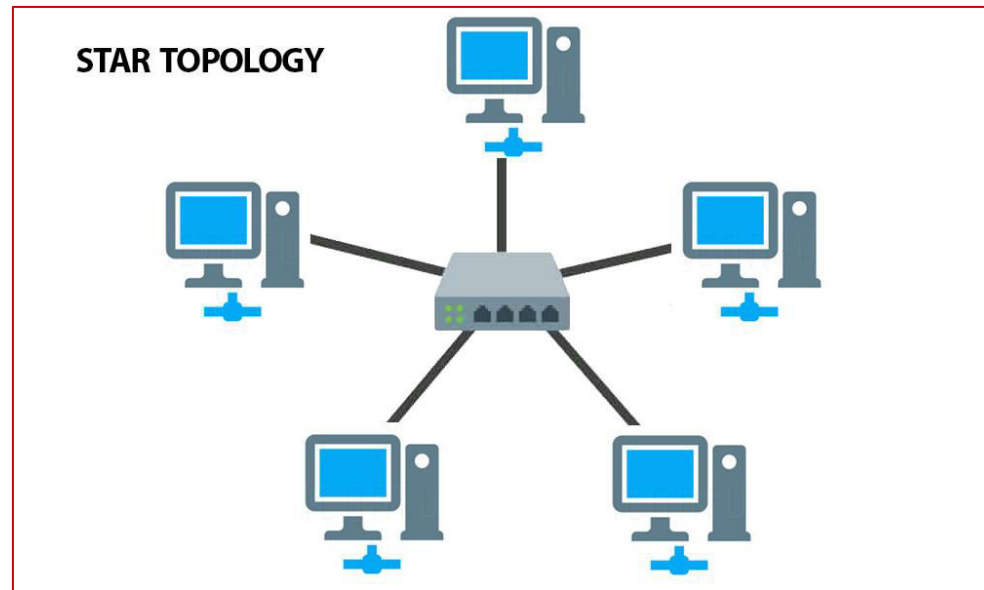
## ☐ **Disadvantages:**

- ☐ **Single Break Causes Trouble:** If one part breaks, the whole network can stop working.
- ☐ **Adding or Removing Devices Is Tricky:** It's hard to change the network by adding or removing devices without disruption.
- ☐ **Can Be Slow:** Sometimes data has to go through many devices, which can make it slower, especially in big networks.

# Network Topology Types

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- ❑ **Star Topology:**
- ❑ In Star Topology, all the computers are connected to a central device through a cable.
- ❑ Thus forming the shape of a star.
- ❑ Coaxial cables or RJ-45 cables are used to connect the computers.



# Network Topology Types

---

## ☐ **Advantages:**

- ☐ Easy Setup: It's simple to connect devices to a central hub or switch.
- ☐ Reliable: If one device fails, it doesn't affect the others, making it a reliable choice.
- ☐ Isolation: Each device has its own connection, so they don't interfere with each other.

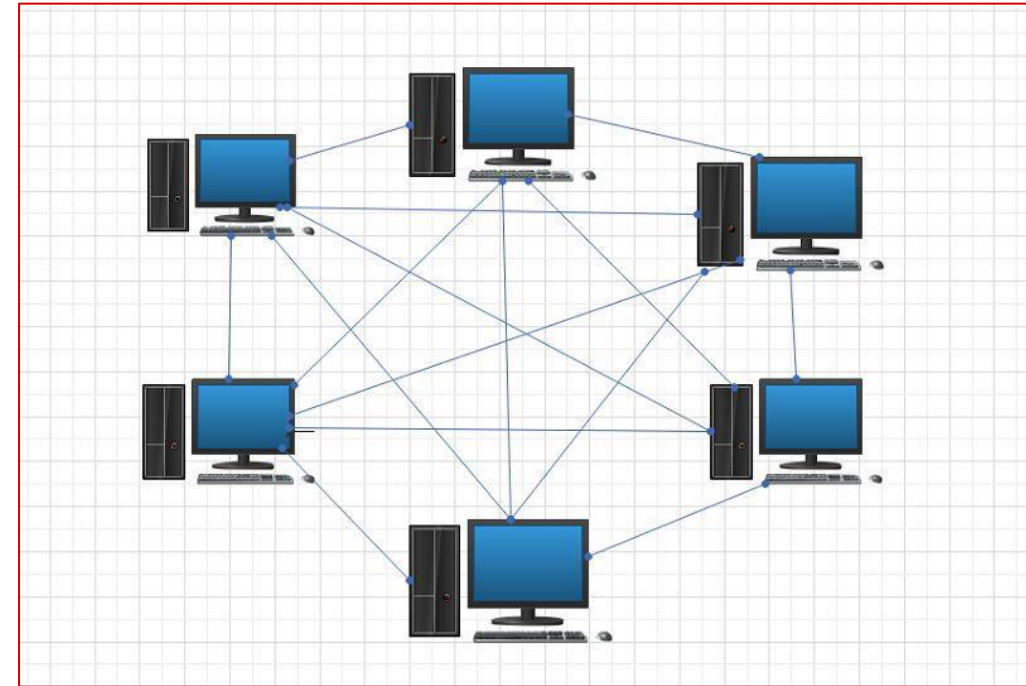
## ☐ **Disadvantages:**

- ☐ Dependence on Central Hub: If the central hub or switch fails, the whole network can go down.
- ☐ Cost: It can be more expensive due to the need for the central hub and separate cables for each device.
- ☐ Limited Scalability: Expanding the network may require additional ports on the central hub, which can be limiting.



# Network Topology Types

- ❑ **Mesh Topology:**
- ❑ In a mesh topology, every device is connected to every other device in the network.
- ❑ This results in a high level of redundancy and fault tolerance.
- ❑ Mesh networks are often used in critical applications where network reliability is paramount.
- ❑ But these networks can be expensive and complex to set up.
- ❑ Use **formula**  $N(N-1)/2$  to find the total number of links in a mesh network.



# Network Topology Types

---

## ☐ **Advantages:**

- ☐ **Reliable:** Mesh topology is very reliable because there are alternative paths for data to travel.
- ☐ **High Redundancy:** Redundancy is built-in, so there are multiple ways for data to reach its destination.
- ☐ **Scalability:** It's easy to add more devices to the network without affecting its overall performance or reliability.

## ☐ **Disadvantages:**

- ☐ **Costly:** The redundancy and multiple connections make it expensive to implement.
- ☐ **Complex to Set Up:** Setting up and managing the numerous connections in a large mesh network can be complex and time-consuming.
- ☐ **Maintenance Challenges:** Troubleshooting and maintaining a mesh network can be challenging due to its complexity.

# Any Question?

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# Thank you

# Information and Communication Technology

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Lecture : 09

Department of Computer Science  
GPGC SWABI (Khyber Pakhtunkhwa)

Fall 2025 Semester  
BS Course: ICT

Course Instructor :  
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# Contents

---

- ☐ The Internet
  - ☐ World Wide Web (WWW)
  - ☐ Web browser
  - ☐ Services and Issues of Internet
- ☐ Website
  - ☐ Types of Websites
  - ☐ Search Engines, Social Media, E-Commerce, Educational Websites
- ☐ Collaborative Computing
- ☐ Social Networking

# The Internet

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- ❑ The internet, short for "interconnected networks," is a global network of computer systems and infrastructure.
- ❑ It allows people and devices all over the world to connect and communicate with each other.
- ❑ It is one of the most significant technological advancements of the late 20th century.
- ❑ The internet has had a profound impact on society, communication, commerce, education, entertainment, and more.
- ❑ It continues to evolve rapidly, with emerging technologies such as 5G and artificial intelligence.

# The Internet

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- ❑ **World Wide Web(WWW):**
- ❑ It is a subset of the internet. The World Wide Web is a system of interconnected web pages and websites that are accessed through the internet using web browsers.
- ❑ **Web Browsers:**
- ❑ Web browsers, such as Google Chrome, Mozilla Firefox, Microsoft Edge, and Safari, are software applications that allow users to access and view web pages. Users enter URLs or use search engines to find specific websites using web browsers.

# The Internet

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## ❑ **Services of Internet:**

- ❑ **Email:** Email (electronic mail) is one of the most widely used internet services, allowing people to send and receive messages electronically. Popular email services include Gmail, Outlook, Yahoo Mail, and many others.
- ❑ **Search Engines:** Search engines like Google, Bing, and Yahoo help users find information on the internet by indexing and organizing web content.
- ❑ **Social Media:** Social networking platforms like Facebook, Twitter, Instagram, and LinkedIn allow users to connect with others, share content, and engage in online communities.
- ❑ **E-Commerce:** Online shopping platforms such as Amazon, eBay, Alibaba, and Shopify allow users to browse, purchase, and sell products and services over the internet.
- ❑ **Cloud Storage:** Cloud storage services like Google Drive, Dropbox, and Microsoft OneDrive provide secure and convenient storage for files and documents accessible from anywhere with an internet connection.



# The Internet

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## ❑ Issues of Internet:

- ❑ **Privacy Concerns:** The collection and misuse of personal data by companies and governments have raised significant privacy concerns. Issues include data breaches, tracking, and profiling.
- ❑ **Cybersecurity Threats:** Scammers use deceptive tactics to manipulate users into revealing sensitive information or performing harmful actions.
- ❑ **Fake News and Misinformation:** The rapid spread of false or misleading information on the internet has the potential to deceive the public and impact elections, public health, and more.
- ❑ **Ethical Issues in Technology:** Emerging technologies like artificial intelligence (AI), facial recognition, and biometrics raise ethical questions about surveillance, bias, and privacy.
- ❑ **Online Addiction and Screen Time:** Excessive screen time and internet addiction can have adverse effects on mental and physical health, particularly in children.

# Website

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- ❑ A website is a collection of related web pages.
- ❑ It is accessed through the internet.
- ❑ A webpage can contain text, audio, video, images, etc.
- ❑ They can serve various purposes, including informational, e-commerce, educational, social networking, and more.

# Types of Websites

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## ☐ **Search Engines:**

- ☐ Search engines are web-based tools that help users find information on the internet such as Google and Bing.
- ☐ They work by indexing and organizing web content, allowing users to search for specific keywords or phrases.
- ☐ Search results are presented as a list of web pages.

# Types of Websites

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## ☐ **Social Media:**

- ☐ Social media platforms, like Facebook, Twitter, and Instagram, enable users to create profiles, connect with others, and share content.
- ☐ Users can post text, images, videos, and links, and engage with content through likes, comments, and shares.
- ☐ Social media has become a key tool for communication, networking, and information sharing.

# Types of Websites

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## ☐ **E-Commerce:**

- ☐ E-commerce (electronic commerce) refers to online buying and selling of products and services.
- ☐ E-commerce websites, such as Amazon and eBay, provide virtual marketplaces where users can browse, shop, and make online payments.
- ☐ Features often include product catalogs, shopping carts, secure payment processing, and order tracking.
- ☐ **Types:**
  - ☐ Business-to-Business (B2B)
  - ☐ Business-to-Consumer (B2C)
  - ☐ Consumer-to-Consumer (C2C)

# Types of Websites

---

## ☐ **Educational Websites:**

- ☐ Educational websites offer a wide range of resources for learning and skill development.
- ☐ These sites may provide courses, tutorials, videos, and interactive content on various subjects and skills.
- ☐ Examples include platforms like Coursera, Khan Academy, and edX, which offer online courses and educational materials.

# Collaborative Computing

---

- ❑ **Collaborative Computing** refers to the use of technology to enable multiple users to work together on a shared task or project.
- ❑ This can include things like shared document editing, real-time communication, and other tools that facilitate collaboration.
- ❑ Collaborative computing is often used in:
  - workplaces,
  - other environments where multiple people need to work together on a common goal.

# Collaborative Computing

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- ❑ Additionally, collaborative computing can also refer to the use of distributed computing resources, where a large task is divided among multiple computers that work together to complete it.
- ❑ There are several benefits to using collaborative computing:
- ❑ **Increased productivity**
- ❑ **Improved communication**
- ❑ **Greater flexibility**
- ❑ **Better decision making**
- ❑ **Cost savings**
- ❑ **Better access to information**



# Social Networking

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- ❑ **Social Networking** refers to the use of digital platforms or websites that allow individuals to connect and share information with each other.
- ❑ The most popular examples of social networking sites include Facebook, Twitter, LinkedIn, and Instagram.
- ❑ These platforms allow users to create a profile, share content, and connect with other users through various features such as messaging, groups, and pages.

# Social Networking

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- ☐ There are several benefits of using social networking:
- ☐ **Connectivity**
- ☐ **Access to information**
- ☐ **Marketing and promotion**
- ☐ **Community building**
- ☐ **Increased opportunities**
- ☐ **Personal expression**

# Social Networking

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- ❑ However, it also has its own set of negative impacts, like:
- ❑ **Addiction to social media**
- ❑ **Cyber bullying**
- ❑ **Privacy concern**
- ❑ **Misinformation**
- ❑ **Lack of face-to-face communication**

# Any Question?

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# Thank you

# Information and Communication Technology

---

Lecture : 10

Department of Computer Science  
GPGC SWABI (Khyber Pakhtunkhwa)

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# Contents

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- ☐ Malware
  - ☐ Types of Malwares
- ☐ Anti-virus
- ☐ Anti-virus vs Anti-malware

# Malware

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- ❑ Malware is short for malicious software and refers to any software that is designed to cause harm to computer systems, networks, or users.
- ❑ Malware can take many forms.
- ❑ It's important for individuals and organizations to be aware of the different types of malware and take steps to protect their systems.

# Malware

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## ☐ **Types of malwares:**

- ☐ Viruses
- ☐ Worms
- ☐ Trojan Horse
- ☐ Ransomware
- ☐ Spyware
- ☐ Key loggers
- ☐ Adware



# Viruses

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- ❑ A Virus is a malicious executable code.
- ❑ The virus spreads when an infected file is passed from one system to another.
- ❑ Viruses can harm your computer by modifying or deleting data.
- ❑ Once a program virus is active, it will infect other programs on the computer.
- ❑ Examples:
  - ❑ Trickbot
  - ❑ WannaCry
  - ❑ Melissa

# Worms

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- ❑ Computer worms are self-replicating programs that spread on their own through computer networks.
- ❑ They enter computers through security holes.
- ❑ Once inside, they make copies and infect other vulnerable systems.
- ❑ They can do harmful things like steal data or damage files.
- ❑ To stay safe, keep your software updated, use antivirus programs, and be careful with suspicious emails or attachments.

# Trojans (Trojan Horses)

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- ❑ Trojans are programs/software that pretend to be doing good but are actually harmful.
- ❑ They can steal data, destroy data, or let hackers control your computer.
- ❑ Trojans don't spread by themselves; It comes packed with another software that you download over the internet.
- ❑ They often come in emails or fake software downloads.
- ❑ To stay safe, use good antivirus software, be careful with downloads, and keep your computer up to date.

# Ransomware

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- ❑ Ransomware is malicious software that locks up your computer or files.
- ❑ It demands a ransom (money) in exchange for unlocking your data.
- ❑ Ransomware often spreads through email attachments or malicious websites.
- ❑ To avoid ransomware, be cautious with email links and downloads, and keep your software updated.
- ❑ Regularly back up your important data to avoid losing it to ransomware.

# Spyware

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- ❑ Spyware is like a secret spy on your computer, quietly watching what you do.
- ❑ It collects your personal information, browsing habits, and more without your permission.
- ❑ Spyware often sneaks into your computer along with other software or through malicious websites.
- ❑ It can slow down your computer and cause unexpected pop-up ads.
- ❑ To protect against spyware, use reputable antivirus software, avoid suspicious downloads, and be cautious about the websites you visit.

# Key Loggers

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- ❑ Keyloggers are like silent spies that record every keystroke you make on your computer.
- ❑ They capture passwords, messages, and sensitive information you type, often without your knowledge.
- ❑ Keyloggers can be hidden in malicious software or disguised as harmless programs.
- ❑ To guard against keyloggers, use reliable antivirus software, avoid suspicious downloads, and be cautious when clicking on links or opening attachments.

# Adware

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- ❑ Adware is software that displays unwanted advertisements, like pop-ups and banners.
- ❑ These ads can be annoying and distracting while you're using your computer.
- ❑ Adware often comes bundled with free software, and users may unknowingly install it.
- ❑ Adware can track your online behavior and collect information about your browsing habits to deliver targeted ads.
- ❑ To avoid adware, be cautious about the software you download, use reputable antivirus software, and regularly scan your computer for unwanted adware.

# Antivirus

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- ❑ Antivirus (or anti-virus) software is a type of cybersecurity tool designed to detect, prevent, and remove malicious software from computer systems and networks.
- ❑ Antivirus software is a crucial component of computer security, and it helps protect your devices from a wide range of threats.
- ❑ Key features and functions of antivirus software:
  - ❑ Virus Detection and protection
  - ❑ Real-Time Protection
  - ❑ Firewall
  - ❑ Email Scanning
  - ❑ Scheduled scans



# Antivirus

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- ☐ Some commonly used antiviruses are:
  - ☐ Norton Antivirus
  - ☐ McAfee Antivirus
  - ☐ Avast Antivirus
  - ☐ AVG Antivirus
  - ☐ Windows Defender

# Antivirus vs Antimalware

| Antivirus                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Antimalware                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <ul style="list-style-type: none"><li>■ Antivirus software primarily focuses on detecting, preventing, and removing computer viruses.</li><li>■ Antivirus software is typically more specialized in dealing with viruses and often provides real-time protection against them.</li><li>■ While many modern antivirus programs have expanded to include broader protection against various types of malware, their primary focus remains on viruses.</li></ul> | <ul style="list-style-type: none"><li>■ Antimalware software, on the other hand, is a broader category of software that provides comprehensive protection against a variety of threats.</li><li>■ Antimalware software is more inclusive and versatile, as it's designed to detect and remove a broader spectrum of malicious software, not just viruses.</li><li>■ Many antimalware programs are equipped to handle not only known threats but also emerging and evolving malware.</li></ul> |

# Any Question?

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# Thank you

# Information and Communication Technology

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Lecture : 11

Department of Computer Science  
GPGC SWABI (Khyber Pakhtunkhwa)

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# Contents

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- ❑ Introduction to Programming Languages
  - ❑ Low level Languages
  - ❑ Assembly Language
  - ❑ High Level Languages
- ❑ Database
  - ❑ Database Management System
  - ❑ Advantages/Disadvantages of DBMS

# Programming Languages

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- ❑ Programming languages are tools that humans use to communicate with computers.
- ❑ These languages consist of a set of rules and instructions that help developers write software and applications.
- ❑ They provide a way to give the computer specific tasks to perform.
- ❑ There are basically two types of computer languages:
  - ❑ Low Level Languages
  - ❑ High Level Languages

# Programming Languages

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- ❑ **Low-Level Languages:**
- ❑ Low-level languages are programming languages that are closely related to the hardware of a computer.
- ❑ This makes it challenging for humans to work on them directly.
- ❑ Examples of low-level languages include:
  - ❑ Machine Language
  - ❑ Assembly Language

# Programming Languages

---

- ❑ **Machine Language:** Machine language consists of binary code, which is a series of 0s and 1s representing specific instructions that a computer's CPU can execute.
- ❑ **Assembly Language:** Assembly language is a more human-readable form of machine language. It uses symbolic instructions, making it easier for programmers to work with, but it's still very closely tied to the computer's architecture.



# Programming Languages

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- ❑ **High-Level Languages:**
- ❑ High-level languages are programming languages designed to be more user-friendly and abstracted from hardware details.
- ❑ They make it easier for programmers to write and understand code.
- ❑ Examples of high-level languages include:
- ❑ C, C++, Java, Python, and more.

# Database

---

- ❑ **Database:**
- ❑ A database is a structured collection of data that is organized and stored in a way that makes it easy to access, manage, and update.
- ❑ Databases store information efficiently and provide a mechanism to retrieve specific pieces of data when needed.
- ❑ Examples of databases include customer records in a company, a library's book catalog, or an online shopping website's product inventory.

# Database

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- ❑ **Database Management System (DBMS):**
- ❑ A Database Management System (DBMS) is software that acts as an intermediary between users and the database.
- ❑ It provides tools and mechanisms to create, manage, retrieve, and secure data stored in a database.
- ❑ DBMS software ensures data integrity, consistency, and security.
- ❑ Popular DBMS examples include:
  - ❑ Microsoft SQL Server, MySQL, Oracle Database and more.

# Database

---

- ❑ **Advantages of DBMS:**
- ❑ **Data Consistency:** DBMS enforces consistency rules, ensuring data accuracy and preventing conflicting information.
- ❑ **Efficient Data Retrieval:** Users can retrieve specific data quickly and easily using query languages like SQL.
- ❑ **Concurrent Access:** DBMS allows multiple users to access and modify the data simultaneously without data corruption.
- ❑ **Data Security:** DBMS offers security features like user access control and encryption.
- ❑ **Data Backup and Recovery:** DBMS provides mechanisms for data backup and recovery in case of data loss.

# Database

---

## ☐ **Disadvantages of DBMS:**

- ☐ Complexity: Setting up and managing a DBMS can be complex, requiring expertise and training.
- ☐ Cost: High-quality DBMS software and the necessary hardware can be expensive, especially for large-scale systems.
- ☐ Performance Overhead: DBMS adds some overhead to data access and manipulation compared to direct file-based data management.

# Any Question?

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# Thank you

# Information and Communication Technology

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GPGC SWABI (Khyber Pakhtunkhwa)

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# Contents

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- ☐ Numbers System
  - ☐ Decimal Number System
  - ☐ Binary Number System
  - ☐ Octal Number System
  - ☐ Hexadecimal Number System



# Numbers

---

- ❑ The arithmetic value which is used for representing the quantity and used in making calculations are defined as NUMBERS.
- ❑ A symbol like “4, 5, 6” which represents a number is known as **numerals**.
- ❑ Without numbers, counting things is not possible, date, time, money, etc.

# Number System

---

- ❑ A number system, also known as a numeral system or base, is a mathematical notation for representing numbers using a set of symbols or digits.
- ❑ There are four commonly used number systems:
  - ❑ **Decimal Number System**
  - ❑ **Binary Number System**
  - ❑ **Octal Number System**
  - ❑ **Hexadecimal Number System**

# Decimal Number System

- ❑ Number system with a base value of 10 is termed a Decimal number system.
- ❑ It uses 10 digits i.e. 0-9 for the creation of numbers.
- ❑ Here, each digit in the number is at a specific place with place value a product of different powers of 10.
- ❑ First place value is called units, second to the left is Tens, so on Hundreds, Thousands, etc.

|     |             |             |              |             |              |             |
|-----|-------------|-------------|--------------|-------------|--------------|-------------|
| ... | 100,000     | 10,000      | 1,000        | 100         | 10           | 1           |
| ... | $10^5$      | $10^4$      | $10^3$       | $10^2$      | $10^1$       | $10^0$      |
| ... | 6           | 5           | 4            | 3           | 2            | 1           |
|     | Sixth digit | Fifth digit | Fourth digit | Third digit | Second digit | First digit |

# Decimal Number System

---

❑ Here, units have the place value as  $10^0$ , tens have the place value as  $10^1$ , hundreds as  $10^2$ , thousands as  $10^3$ , and so on.

❑ For example, 10264 has place values as,  
 $= (1 \times 10^4) + (0 \times 10^3) + (2 \times 10^2) + (6 \times 10^1) + (4 \times 10^0)$

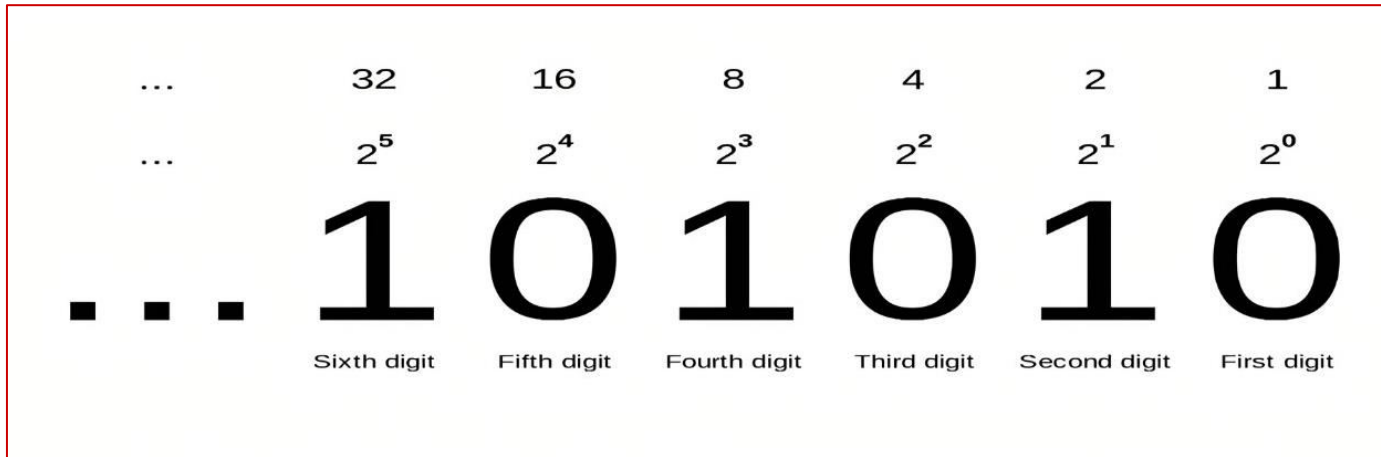
$$= (1 \times 10000) + (0 \times 1000) + (2 \times 100) + (6 \times 10) + (4 \times 1)$$

$$= 10000 + 0 + 200 + 60 + 4$$

$$= 10264$$

# Binary Number System

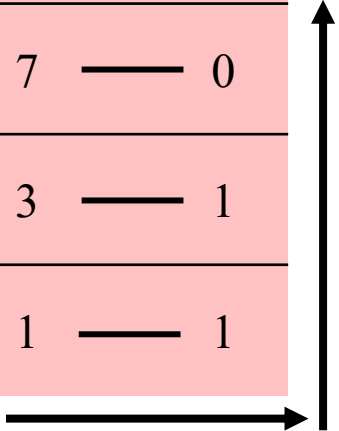
- ❑ Number System with base value 2 is termed as Binary number system.
- ❑ It uses 2 digits i.e. 0 and 1 for the creation of numbers.
- ❑ The numbers formed using these two digits are termed Binary Numbers.
- ❑ The binary number system is used in electronic devices and computer systems to represent the two states ON and OFF.



# Binary Number System

- ❑ **Decimal To Binary Conversion:**
- ❑ Recursively divide the decimal number by 2.
- ❑ If there is a remainder write it separately.
- ❑ When the decimal number reaches to 1 stop the division.
- ❑ Write the remainders of each step in reverse order which will give you the binary equivalent.
- ❑ E.g. Convert the decimal number 14 to its binary equivalent.
- ❑ **Binary Equivalent : 1110**

|   |       |
|---|-------|
| 2 | 14    |
| 2 | 7 — 0 |
| 2 | 3 — 1 |
|   | 1 — 1 |



# Binary Number System

---

- ❑ **Binary To Decimal Conversion:**
- ❑ Multiply each bit with its place values.
- ❑ Add all the numbers, this will give you the decimal equivalent.
- ❑ For example, the binary number (11001) in decimal would be equal to 25.  
$$= (1 \times 2^4) + (1 \times 2^3) + (0 \times 2^2) + (0 \times 2^1) + (1 \times 2^0)$$
$$= (1 \times 16) + (1 \times 8) + (0 \times 4) + (0 \times 2) + (1 \times 1)$$
$$= 16 + 8 + 0 + 0 + 1$$
$$= 25$$

# Octal Number System

---

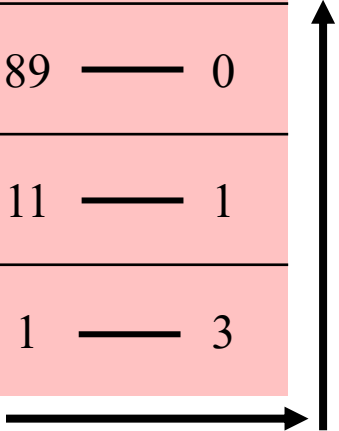
- ❑ Octal Number System is one in which the base value is 8.
- ❑ It uses 8 digits i.e. 0-7 for the creation of Octal Numbers.
- ❑ Here the place values are  $8^0$ ,  $8^1$ , and  $8^2$ .
- ❑ Octal Numbers are useful for the representation of UTF8 Numbers.



# Octal Number System

- ❑ **Decimal To Octal Conversion:**
- ❑ Recursively divide the decimal number by 8.
- ❑ If there is a remainder write it separately.
- ❑ When the decimal number becomes less than 8 stop the division.
- ❑ Write the remainders of each step in reverse order which will give you the octal equivalent.
- ❑ E.g. Convert the decimal number 712 to its octal equivalent.
- ❑ **Octal Equivalent : 1310**

|   |        |
|---|--------|
| 8 | 712    |
| 8 | 89 — 0 |
| 8 | 11 — 1 |
|   | 1 — 3  |



# Octal Number System

---

- ❑ **Octal To Decimal Conversion:**
- ❑ Multiply each digit with its place values.
- ❑ Add all the numbers, this will give you the decimal equivalent.
- ❑ For example, the octal number 1731 in decimal would be equal to 985.  
$$= (1 \times 8^3) + (7 \times 8^2) + (3 \times 8^1) + (1 \times 8^0)$$
$$= (1 \times 512) + (7 \times 64) + (3 \times 8) + (1 \times 1)$$
$$= 512 + 448 + 24 + 1$$
$$= 985$$

# Hexadecimal Number System

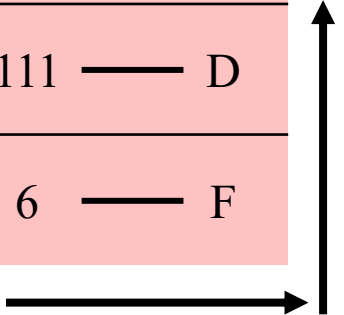
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- ❑ Number System with base value 16 is termed as Hexadecimal Number System.
- ❑ It uses 16 digits for the creation of its numbers.
- ❑ Digits from 0-9 are taken like the digits in the decimal number system but the digits from 10-15 are represented as A-F i.e. 10 is represented as A, 11 as B, 12 as C, 13 as D, 14 as E, and 15 as F.
- ❑ Hexadecimal Numbers are useful for handling memory address locations.

# Hexadecimal Number System

- ❑ **Decimal To Hexadecimal Conversion:**
- ❑ Recursively divide the decimal number by the number 16/F.
- ❑ If there is a remainder write it separately.
- ❑ When the decimal number becomes less than 16/F stop the division.
- ❑ Write the remainders of each step in reverse order which will give you the hexadecimal equivalent.
- ❑ E.g. Convert the decimal number 1789 to its hexadecimal equivalent.
- ❑ **Hexadecimal Equivalent : 6FD**

|    |         |
|----|---------|
| 16 | 1789    |
| 16 | 111 — D |
|    | 6 — F   |



# Hexadecimal Number System

---

- ❑ **Hexadecimal To Decimal Conversion:**
- ❑ Multiply each digit with its place values.
- ❑ Add all the numbers, this will give you the decimal equivalent.
- ❑ For example, the hexadecimal number 9E1B in decimal would be equal to 40475.

$$= (9 \times 16^3) + (E \times 16^2) + (1 \times 16^1) + (B \times 16^0)$$

$$= (9 \times 4096) + (14 \times 256) + (1 \times 16) + (11 \times 1)$$

$$= 36864 + 3584 + 16 + 11$$

$$= 40475$$

# Any Question?

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# Thank you

# Information and Communication Technology

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Lecture : 13

Department of Computer Science  
GPGC SWABI (Khyber Pakhtunkhwa)

Fall 2025 Semester  
BS Course: ICT

Course Instructor :  
Mr. Ihtisham Ul Haq  
Lecturer (CS)  
GPGC Swabi

# Contents

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- ☐ MS Word
  - ☐ Introduction
  - ☐ MS Word (Interface)
  - ☐ Creating and Saving Documents
  - ☐ Editing and Formatting Documents
  - ☐ Printing Documents
  - ☐ MS Word Shortcut Keys



# MS Word

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- ❑ MS Word is a part of the MS Office package provided by Microsoft.
- ❑ Microsoft Word is a powerful word processing software developed by Microsoft, widely used for creating and editing text documents.
- ❑ It is used for generating letters, reports, applications, meeting agendas, calendar, Resumes, Memorandums, newspaper articles, business cards and many other documents.
- ❑ It provides an easy interface to work with different documents.

# MS Word (Versions)

---

## ☐ **MS Word Versions:**

- ☐ MS Word (1994)
- ☐ MS Word (1995)
- ☐ MS Word (1997)
- ☐ MS Word (2000)
- ☐ MS Word (2003)
- ☐ MS Word (2007)
- ☐ MS Word (2010)
- ☐ MS Word (2013)
- ☐ MS Word (2016)
- ☐ MS Word (2019)
- ☐ MS Word 365

# MS Word (Extensions)

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- ☐ The extension name of MS word 2003 or earlier version is .doc.
- ☐ The extension name of MS word 2007 onwards all version are .docx.

# MS Word (Features)

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- ❑ Easy table creation and formatting features such as table auto format, which allows you to focus on and let word handle that formatting.
- ❑ Auto correct and auto format feature that catch typographical errors automatically and allow you to use predefined shortcuts and typing patterns to quickly format a document.
- ❑ Large document management feature, like the ability to create table of contents, index and list of figure.
- ❑ Automation by means of macro creation.

# MS Word (How to open?)

---

- ☐ How to open MS Word in Windows 10 and newer versions.

## 1) From the Start Menu:

- ☐ Click the Start Button at the left bottom of your screen to show the Start Menu.
- ☐ Scroll the all programs section.
- ☐ Choose MS Word.

## 2) From the Search bar:

- ☐ Click on the Search Bar in the taskbar and search for Word by typing Word.
- ☐ Click on MS Word.

# MS Word (How to open?)

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3) From right click menu:

- ☐ Right-click on any empty place
- ☐ Select New from the menu
- ☐ Then select MS Word Document from the list

# MS Word (New Document)

---

- ☐ **Create a New Document**

- ☐ There are two ways to create new documents:

1)

- ☐ Click the File Tab and Click New.
- ☐ Then click blank document or any other template as your like.

2)

- ☐ Press CTRL+N on the keyboard .
- ☐ It will open a new Word document.

- ☐ **Note:**

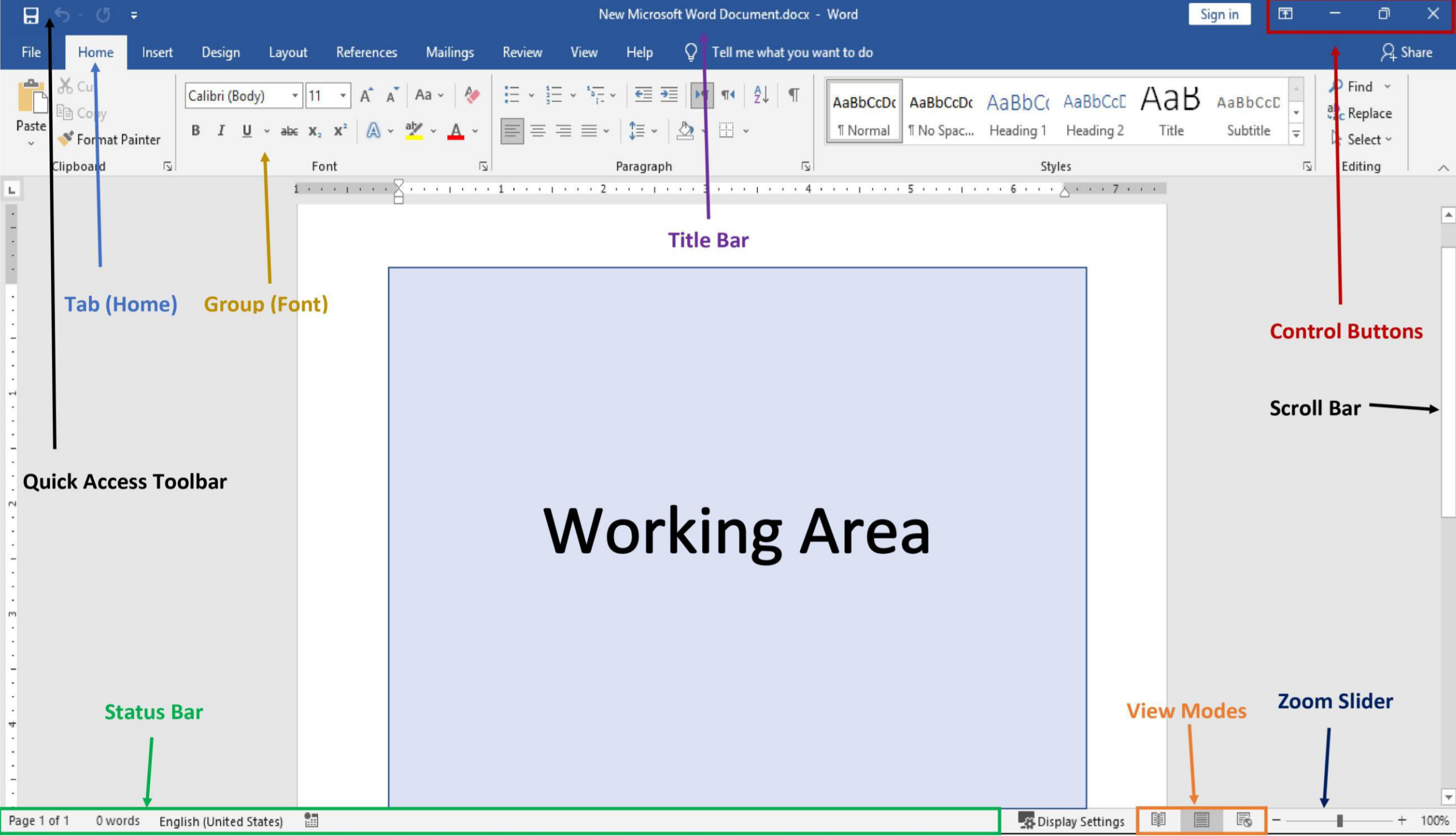
- ☐ When you click on the File tab and Click New, you have many choices about the types of documents you can create. If you wish to start from a blank document, click Blank. If you wish to start from a template, you can browse through your choices on the left, see the choices on center screen, and preview the selection on the right screen.

# MS Word (Interface)

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- ❑ MS Word provides an easy to use interface to work with different types of documents.





# MS Word (Interface)

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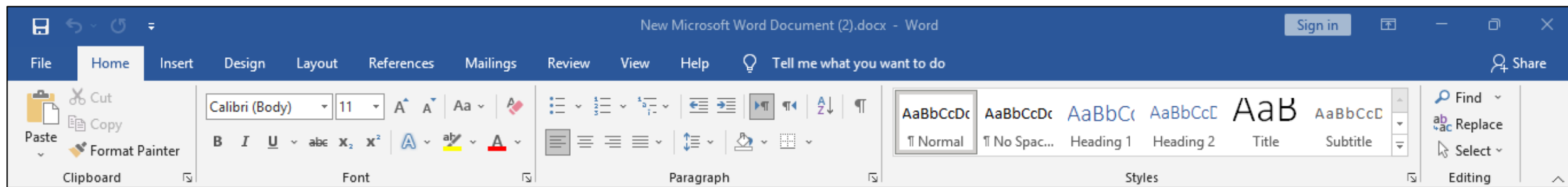
- ❑ **Quick Access Toolbar:**
- ❑ The quick access toolbar is a customizable toolbar that contains commands that you may want to use.
- ❑ You can place the quick access toolbar above or below the ribbon.
- ❑ You can also add items to the quick access toolbar.



# MS Word (Interface)

---

- ❑ **The Ribbon:**
- ❑ The Ribbon is the panel at the top portion of the document.
- ❑ It has nine tabs: File, Home, Insert, Design, Layout, References, Mailings, Review, and View that contain many new and existing features of Word.
- ❑ Each tab is divided into groups.



# MS Word (Interface)

---

- ❑ **File:** Using file menu, one can save the file, close the file, open the file, create new document, print the file, customize the document, print, send (through email or fax), publish or close.
- ❑ **Home:** Clipboard, Fonts, Paragraph, Styles, and Editing.
- ❑ **Insert:** Pages, Tables, Illustrations, Links, Add-ins, Media, Comments, Header & Footer, Text, and Symbols
- ❑ **Design:** Themes, Page Background
- ❑ **Layout:** Page Setup, Paragraph, Arrange
- ❑ **References:** Table of Contents, Footnote, Research, Citation & Bibliography, Captions, Index, and Table of Authorities

# MS Word (Interface)

---

- ❑ **Mailings:** Create, Start Mail Merge, Write & Insert Fields, Preview Results, Finish
- ❑ **Review:** Proofing, Accessibility, Language, Comments, Tracking, Changes, Compare, Protect
- ❑ **View:** Document Views, Page Movement, Show/Hide, Zoom, Window, Macros, SharePoint

# MS Word (Saving document)

## ❑ Saving a Document:

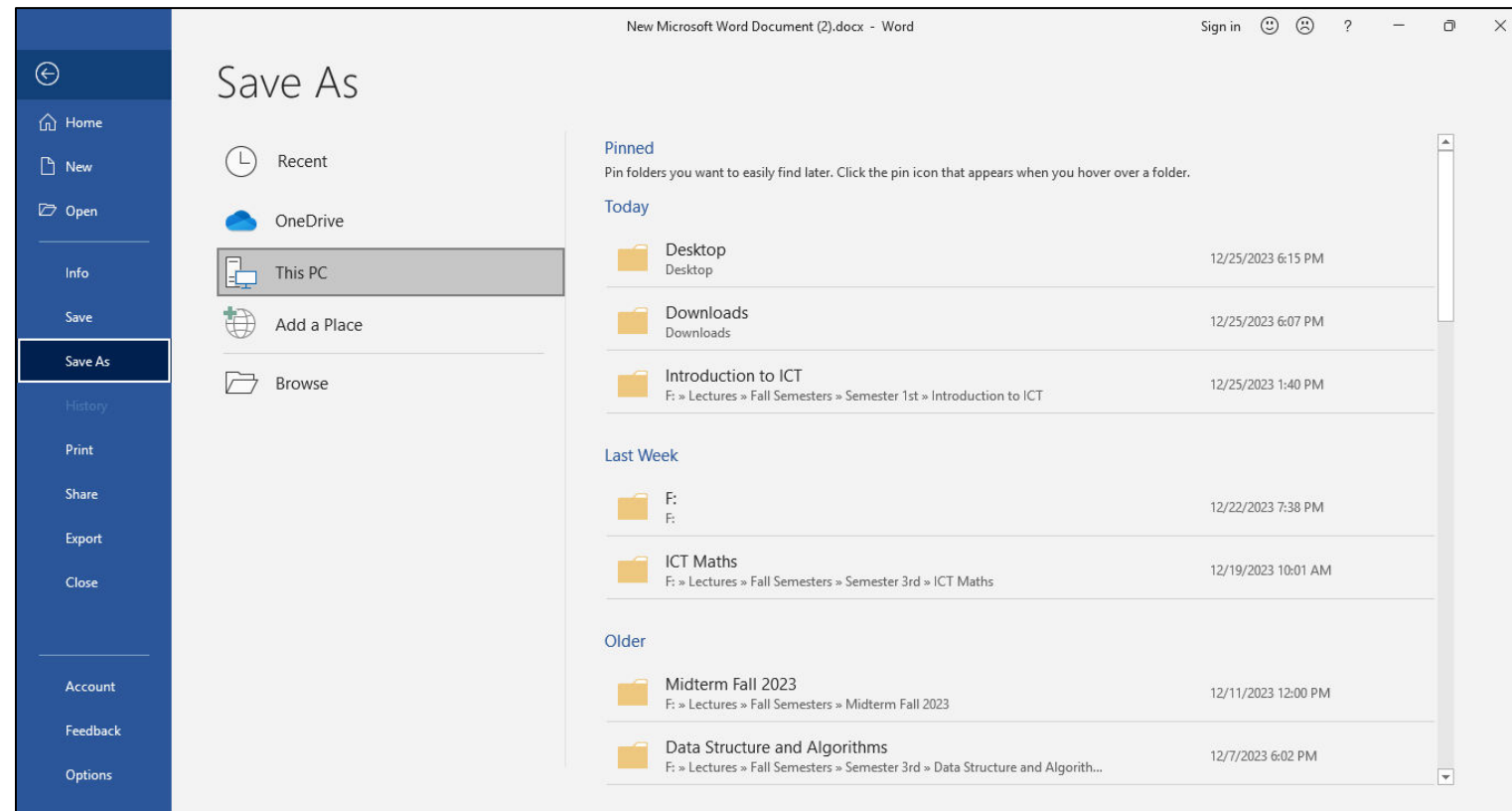
1) Click the File Tab and Click Save or Save As

OR

2) Click on save icon (towards the top-left on the screen)

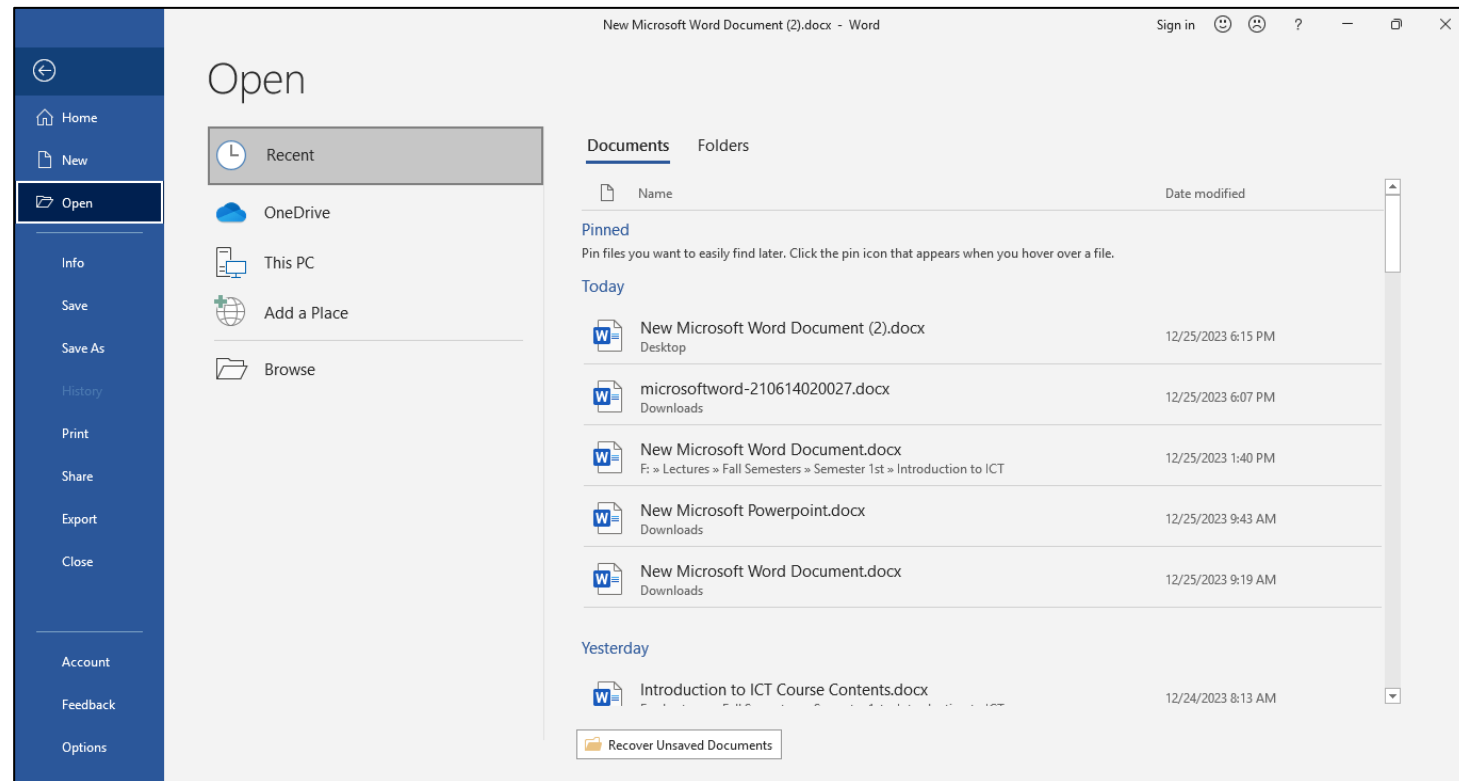
OR

3) Press CTRL+S on the keyboard and the file will be saved



# MS Word (Opening document)

- ❑ **Opening a Document:**
- ❑ Click the File Tab and Click Open,
- ❑ If you have recently used the document, you can select and double click on document from the Right panel recent file list.
- ❑ Then select the file which you want to open from your saved location.



# MS Word (Opening document)

---

- ☐ OR
- ☐ If you want to open the file that is stored in your hard drive then
- ☐ Go to the PC and open the location where you have stored the file.
- ☐ Then select the file which you want to open from your saved location.
- ☐ Click open command button.



# MS Word (Renaming documents)

---

- ❑ **Renaming Documents:**
- ❑ To rename a Word document:
- ❑ Open My Computer and find the file you want to rename.
- ❑ Right-click the document name with the mouse and select Rename from the shortcut menu.
- ❑ Type the new name for the file and press the ENTER key.

# MS Word (Inserting text)

---

- ☐ **Inserting Additional Text:**
- ☐ Text can be inserted in a document at any point using any of the following methods:
- ☐ **Type Text:**
- ☐ Put your cursor where you want to add the text and begin typing.

# MS Word (Cut, Copy, Paste)

---

## ☐ **Copy and Paste Text:**

- ☐ Highlight the text you wish to copy and right click and click Copy, put your cursor where you want the text in the document and right click and click Paste.
- ☐ OR Ctrl + C for copying and Ctrl + V for pasting.

## ☐ **Cut and Paste Text:**

- ☐ Highlight the text you wish to copy, right click, and click Cut, put your cursor where you want the text in the document, right click, and click Paste.
- ☐ OR Ctrl + X for cutting and Ctrl + V for pasting.

# MS Word (Drag & Delete text)

---

## ☐ **Drag Text:**

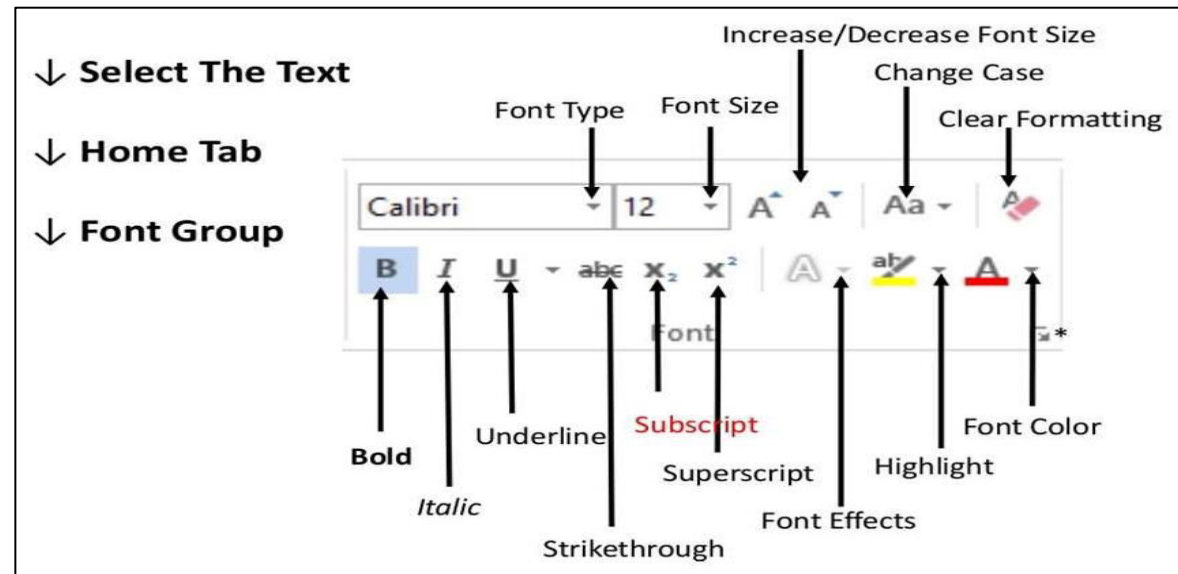
- ☐ Highlight the text you wish to move, click on it and drag it to the place where you want the text in the document.
- ☐ You can also use the Clipboard group on the Ribbon.

## ☐ **Deleting Blocks of Text:**

- ☐ Use the BACKSPACE and DELETE keys on the keyboard to delete text.
- ☐ Backspace will delete text to the left of the cursor and Delete will erase text to the right.
- ☐ To delete a large selection of text, highlight it using any of the methods outlined above and press the DELETE key.

# MS Word (Home Tab)

- ❑ **Font Group:**
- ❑ It allows formatting the text.
- ❑ It has some buttons such as bold, italic, underline, strikethrough, superscript, subscript, change case, font color, text highlight color, grow font, shrink font button, font typeface and font size combo box.



# MS Word (Home Tab)

---

- ☐ **Bold** button allows make the selected text bold.
- ☐ **Italic** button allows italicize the selected text.
- ☐ **Underline** button allows underline the selected text and also can change the underline style and color.
- ☐ **Strikethrough** allows to draw a line through the middle of the selected text.
- ☐ **Subscript** create a small letter below the text baseline.
- ☐ **Superscript** create a small letter above the line of the text.

# MS Word (Home Tab)

---

- ❑ **Change Case** buttons allows change all selected text to uppercase, lowercase or other common capitalizations.
- ❑ When you click on change case button, display some option such as Sentence case, lower case, UPPER CASE, Capitalize Each Word and Toggle Case.
- ❑ **Text highlight color** allows make text look like it was marked with a highlighter pen and can be selected any color from the highlight color list. Suppose you want to remove highlight color of the given highlight text, then choose no option from highlight button.
- ❑ **Font color** list button allows change text color and it display some color.
- ❑ **Text effect and typography** buttons allows add some effect to you text like shadow and grow. If you document has been opened in compatibility mode, this button disabled.

# MS Word (Home Tab)

---

- ☐ **To change the font type face:**
- ☐ First select the text which you want o changes.
- ☐ Click Home tab of the ribbon.
- ☐ Click the arrow next to the font name and choose a font of the font group of the ribbon.
- ☐ Remember that you can preview how the new font will look by highlighting the text, and hovering over the new font typeface.



# MS Word (Home Tab)

---

- ☐ **To change the font size:**
- ☐ Click the arrow next to the font size and choose the appropriate size, or Click the increase or decrease font size buttons.
- ☐ **Font Styles and Effects:**
- ☐ Font styles are predefined formatting options that are used to emphasize text.
- ☐ They include: Bold, Italic, and Underline. To add these to text:
- ☐ Select the text and click the Font Styles included on the Font Group of the Ribbon,
- ☐ OR
- ☐ Select the text and right click to display the font tools

# MS Word (Home Tab)

---

- ☐ **Change Text Color:**
- ☐ Select the text and click the Colors button included on the Font Group of the Ribbon,
- ☐ OR
- ☐ Highlight the text, right click, and choose the colors tool.
- ☐ Select the color by clicking the down arrow next to the font color button.

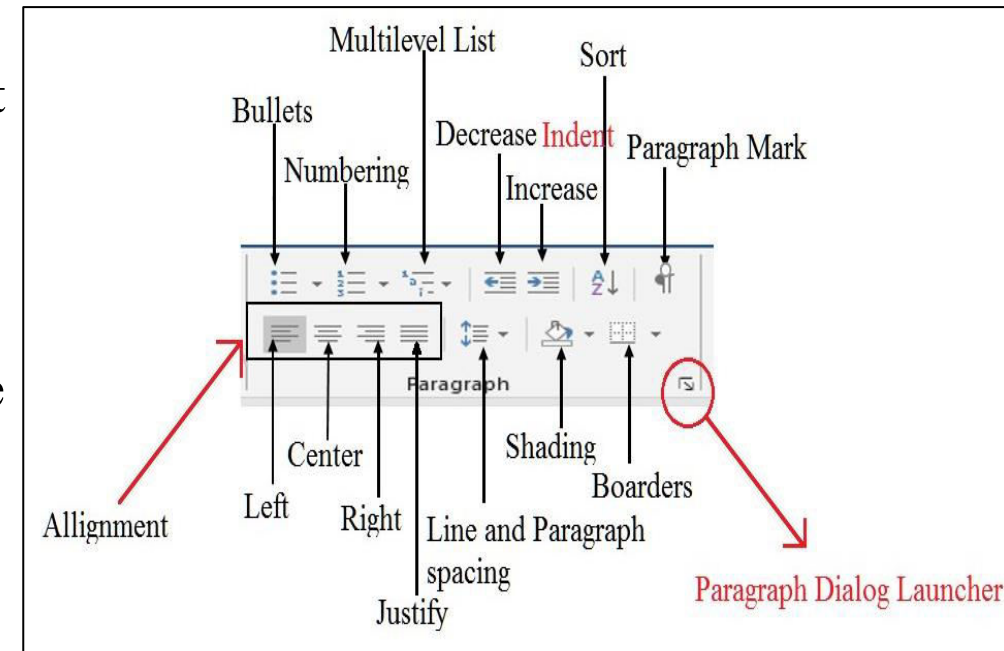
# MS Word (Home Tab)

---

- ❑ **Highlight Text:**
- ❑ Highlighting text allows you to use emphasize text as you would if you had a marker.
- ❑ To highlight text:
  - ❑ Select the text.
    - 1) Click the Highlight Button on the Font Group of the Ribbon,  
OR
    - 2) Select the text and right click and select the highlight tool
  - To change the color of the highlighter, click on down arrow next to the highlight button.

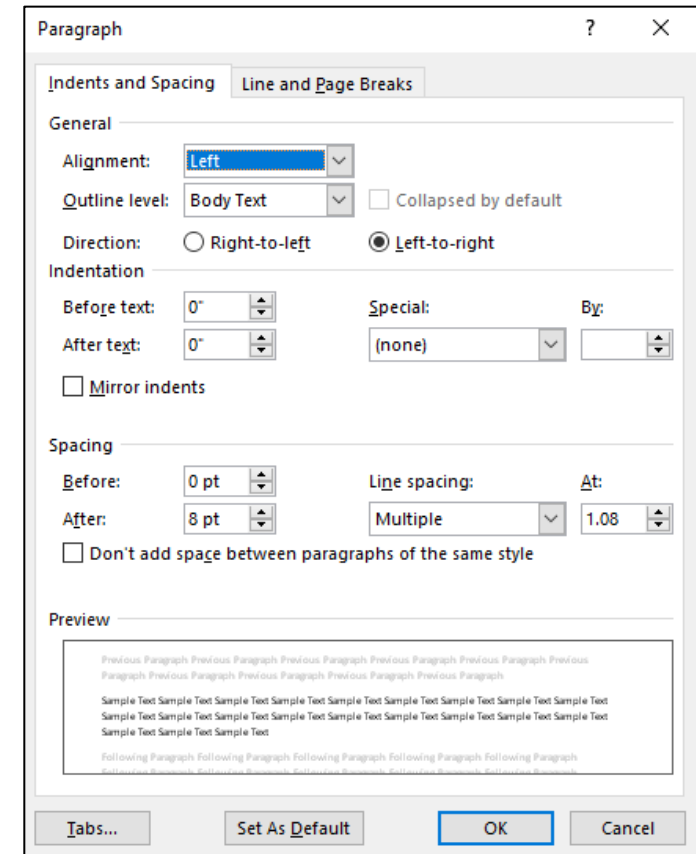
# MS Word (Home Tab)

- ❑ **Paragraph Group:**
- ❑ **Change Paragraph Alignment:**
- ❑ The paragraph alignment allows you to set how you want your text to appear.
- ❑ To change the alignment:
- ❑ Click the Home Tab
- ❑ Choose the appropriate button for alignment on the Paragraph Group.
  - ❑ **Align Left:** the text is aligned with your left margin
  - ❑ **Center:** The text is centered within your margins
  - ❑ **Align Right:** Aligns text with the right margin.
  - ❑ **Justify:** Aligns text to both the left and right margins.



# MS Word (Home Tab)

- ❑ **Indent Paragraphs:**
- ❑ Indenting paragraphs allows you set text within a paragraph at different margins.
- ❑ There are several options for indenting:
  - ❑ **First Line:** Controls the left boundary for the first line of a paragraph.
  - ❑ **Hanging:** Controls the left boundary of every line in a paragraph except the first one.
  - ❑ **Left:** Controls the left boundary for every line in a paragraph.
  - ❑ **Right:** Controls the right boundary for every line in a paragraph.



# MS Word (Home Tab)

---

- ☐ To indent paragraphs, you can do the following:
  - ☐ Click the Indent buttons to control the indent.
  - ☐ Click the Indent button repeated times to increase the size of the indent.
  - ☐ Click the dialog box of the Paragraph Group.
  - ☐ Click the Indents and Spacing Tab.
  - ☐ Select your indents.

# MS Word (Home Tab)

---

- ☐ **Add Borders and Shading:**
- ☐ You can add borders and shading to paragraphs and entire pages. To create a border around a paragraph or paragraphs:
- ☐ Select the area of text where you want the border or shading.
- ☐ Click the Borders Button on the Paragraph Group on the Home Tab
- ☐ Choose the Border and Shading
- ☐ Choose the appropriate options

# MS Word (Home Tab)

---

- ☐ **Change Spacing Between Paragraphs and Lines:**
- ☐ You can change the space between lines and paragraphs by doing the following:
- ☐ Select the paragraph or paragraphs you wish to change.
- ☐ On the Home Tab, Click the Paragraph group launcher button
- ☐ Click the Indents and Spacing Tab
- ☐ In the Spacing section, adjust your spacing accordingly



# MS Word (Home Tab)

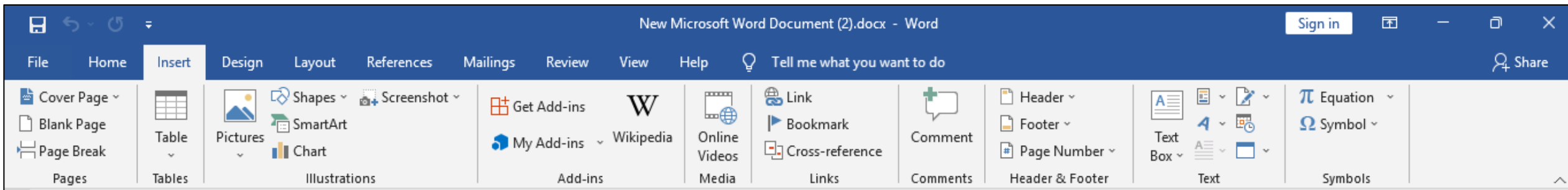
---

- ☐ **Search and Replace Text:**
- ☐ To find a particular word or phrase in a document:
  - ☐ Click Home tab of the ribbon.
  - ☐ Click Find on the Editing Group.
- ☐ To find and replace a word or phrase in the document:
  - ☐ Click Home tab of the ribbon.
  - ☐ Click Replace on the Editing Group of the Ribbon. (Displays the find and replace dialog box)
- ☐ **Note:**
- ☐ Find and replace dialog box contain three tab such as find, replace and Go to.

# MS Word (Insert Tab)

---

- ❑ By using insert Tab, we can insert specific object into current work area.
- ❑ Object means picture, Table, symbol, word art, Date & time, Any symbol, Chart, Page number, blank page, etc.



# MS Word (Insert Tab)

---

- ❑ **Adding Tables:**
- ❑ Tables are used to display data in a table format.
- ❑ Place the cursor on the page where you want the new table
- ❑ Click the Insert Tab of the Ribbon
- ❑ Click the Tables Button on the Tables Group. You can create a table one of four ways:
  - 1) Highlight the number of row and columns.
  - 2) Click Insert Table.
  - 3) Click the Draw Table, create your table by clicking and entering the rows and column.
  - 4) Click Quick Tables and choose tables which are predefined table by the MS Office.

# MS Word (Insert Tab)

---

- ❑ **Symbols and Special Characters:**
- ❑ Special characters are punctuation, spacing, or typographical characters that are not generally available on the standard keyboard. To insert symbols and special characters:
  - ❑ Place your cursor in the document where you want the symbol
  - ❑ Click the Insert Tab on the Ribbon.
  - ❑ You will see the symbol button as illustrated (displayed as right end bottom of the ribbon).
  - ❑ Click the Symbols drop down button will display drop down list of symbols option.
  - ❑ Clicking on any one of these will insert the symbol that you clicked on (at the insertion point).

# MS Word (Insert Tab)

---

- ☐ If Clicking on more symbols will display additional symbols as illustrator.
- ☐ If you click on special character tab within dialog box, display the list of special character.
- ☐ Click on insert button.
- ☐ Now you will see in your document, selected symbol will be inserted.
- ☐ Then after completing insert symbols, click close button.

# MS Word (Insert Tab)

---

- ❑ **Equations:**
- ❑ Word also allows you to insert mathematical equations.
- ❑ allows insert common mathematical equations or build up your own equations using a library of math symbols.
- ❑ To access the mathematical equations tool:
  - ❑ Place your cursor in the document where you want the symbol
  - ❑ Click the Insert Tab on the Ribbon
  - ❑ Click the symbols drop down Button and then choose Equations.
  - ❑ Choose the appropriate equation and structure or click Insert New Equation
  - ❑ To edit the equation, click the equation and the Design Tab will be available in the Ribbon

# MS Word (Insert Tab)

---

- ☐ **Illustrations, Pictures, and Smart Art:**
- ☐ Word allows you to insert illustrations and pictures into a document from online or your saved image.
- ☐ Place your cursor in the document where you want the illustration/picture
- ☐ Click the Insert Tab on the Ribbon
- ☐ Click the any picture button from your illustrations groups according your requirement.

# MS Word (Insert Tab)

---

- ☐ **To insert a picture:**
  - ☐ Place your cursor in the document where you want the /picture
  - ☐ Click the Insert Tab on the Ribbon
  - ☐ Click the Picture Button
  - ☐ Browse to the picture you wish to insert in to your document and select it.
  - ☐ Click Insert

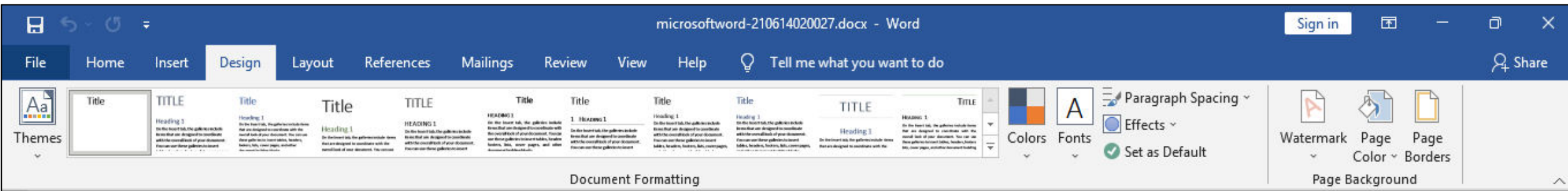


# MS Word (Insert Tab)

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- ❑ **Smart Art:**
- ❑ It is a collection of graphics you can utilize to organize information within your document. It includes timelines, processes, or workflow.
- ❑ **To insert Smart Art:**
  - ❑ Place your cursor in the document where you want the illustration/picture.
  - ❑ Click the Insert Tab on the Ribbon.
  - ❑ Click the Smart Art button.
  - ❑ Click the Smart Art you wish to include in your document.
  - ❑ Click the arrow on the left side of the graphic to insert text or type the text in the graphic.

# MS Word (Design Tab)



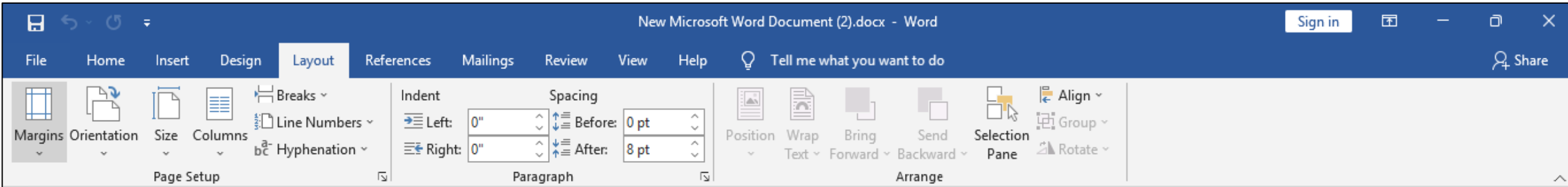
- ☐ **Apply a Page Border and Color:**
- ☐ Click the Design Tab on the Ribbon
- ☐ On the Page Background Group, click the Page Colors or Page Borders drop down menus

# MS Word (Design Tab)

---

- ❑ **Watermarks:**
- ❑ A watermark is a translucent image that appears behind the primary text in a document.  
To insert a watermark:
- ❑ Click the Design Tab in the Ribbon
- ❑ Click the Watermark Button in the Page Background Group
- ❑ Click the Watermark you want for the document or click Custom Watermark and create your own watermark
- ❑ To remove a watermark, follow the steps above, but click Remove Watermark

# MS Word (Layout Tab)



- ☐ **Modify Page Margins:**
- ☐ Click the Layout Tab on the Ribbon
- ☐ On the Page Setup Group, Click Margins.
- ☐ A drop down list margin box will be displayed.
- ☐ As you move the pointer over the drop down list box, different default margin will be applied.
- ☐ Click on a margin as you like to apply it your document.

# MS Word (Layout Tab)

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- ☐ If you want to create your own custom margin, then choose custom margin command displayed at the bottom of margins drop down list box.
- ☐ Display page setup dialog box.
- ☐ Then choose the margin as your like.
- ☐ After completed choose margin, click on ok command button.

# MS Word (Layout Tab)

---

- ☐ **Customized Page Setup:**

- ☐ **To change the Orientation, Size of the Page, or Columns:**

- ☐ Click the Layout Tab on the Ribbon
- ☐ On the Page Setup Group, Click the Orientation, Size, or Columns drop down menus
- ☐ Click the appropriate choice

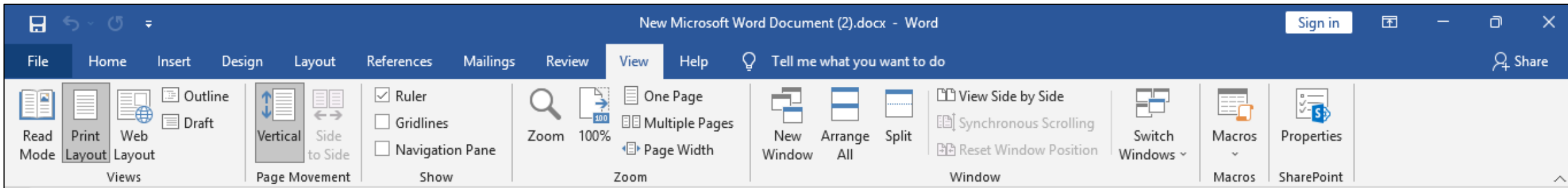
# MS Word (Layout Tab)

---

- ☐ **Size:**
- ☐ To change the paper size for your whole document:
- ☐ On the Layout tab, in the Page Setup group, click Size.
- ☐ Select More Paper Sizes.
- ☐ In the Page Setup dialog box, choose a paper size and for Apply to, select Whole document.
- ☐ Click OK.

# MS Word (View Tab)

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## ☐ **View Modes:**

- ☐ Microsoft Word offers different view modes to help users edit, format, and navigate through documents in various ways. Here are the common view modes in Microsoft Word:



# MS Word (View Tab)

---

## ☐ **Print Layout View:**

- ☐ This is the default view in Word.
- ☐ It shows how the document will appear when printed, including margins, headers, footers, and page breaks.

## ☐ **Read Mode:**

- ☐ This view is optimized for reading documents on the screen.
- ☐ It provides a clean, distraction-free layout for easy reading.

## ☐ **Web Layout View:**

- ☐ This view is designed for creating and viewing documents as they would appear on the web.
- ☐ It adjusts the layout to fit the screen width and displays background colors and images.

# MS Word (View Tab)

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- ☐ **Outline View:**

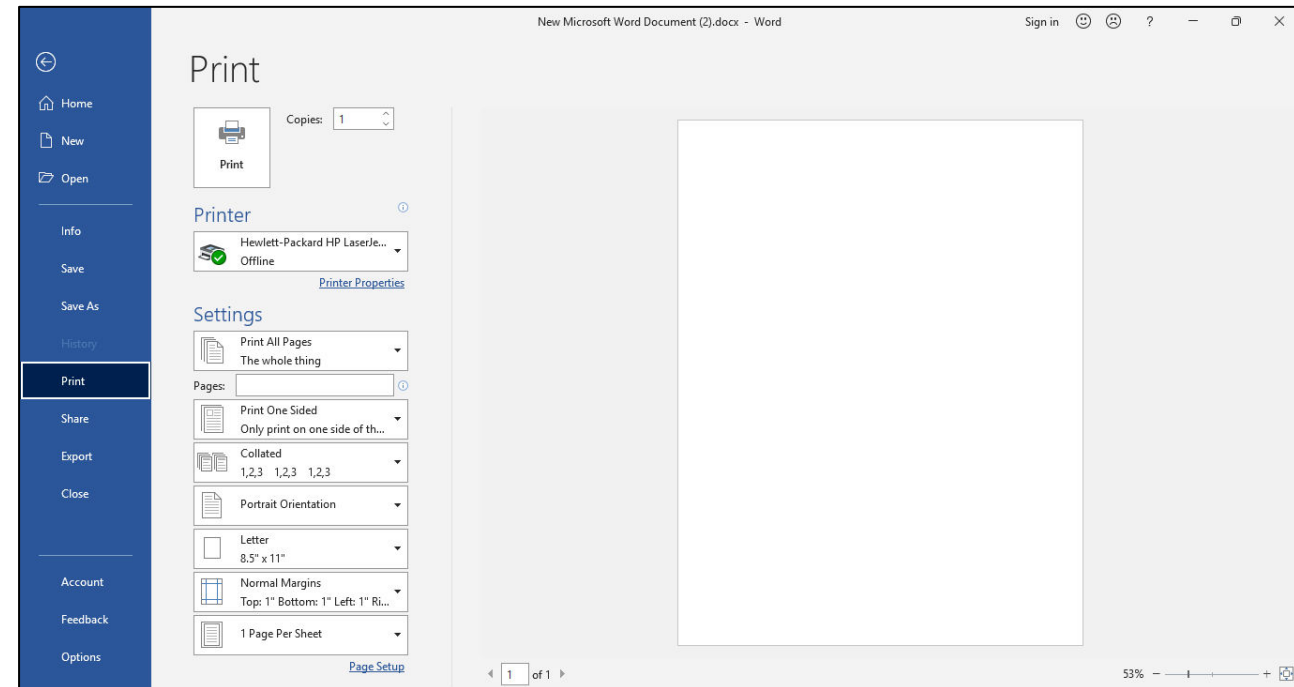
- ☐ This view displays the structure of the document in an outline format.
- ☐ It's useful for organizing and navigating large documents with headings and subheadings.

- ☐ **Draft View:**

- ☐ In Draft View, you see a simplified version of the document without images and formatting.
- ☐ It is often used for quick editing and focusing on the text.
- ☐ To switch between these view modes in Microsoft Word, you can usually find options in the View tab on the Ribbon.

# MS Word (Printing)

- ❑ **Printing Options:**
- ❑ To print a document you can use either of the options:
  - 1) **From the file tab:**
    - ❑ Select the file tab
    - ❑ From the menu, select the Print option
    - ❑ Select the appropriate options in the print dialog box and click Print.
  - 2) **Using shortcut key:**
    - ❑ Press CTRL+P
    - ❑ Select the appropriate options in the print dialog box and click Print.



# MS Word Shortcut Keys

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## ☐ **Editing Shortcuts:**

- ☐ Ctrl + C: Copy selected text.
- ☐ Ctrl + X: Cut selected text.
- ☐ Ctrl + V: Paste copied or cut text.
- ☐ Ctrl + Z: Undo the last action.
- ☐ Ctrl + Y: Redo the last undone action.

## ☐ **Selection Shortcuts:**

- ☐ Shift + Arrow Keys: Extend selection by one character/line.
- ☐ Ctrl + Shift + Arrow Keys: Extend selection by one word.
- ☐ Ctrl + A: Select the entire document.
- ☐ Shift + F8: Activate the extended selection mode. Press again to expand selection.

# MS Word Shortcut Keys

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## ☐ **Formatting Shortcuts:**

- ☐ Ctrl + B: Bold.
- ☐ Ctrl + I: Italic.
- ☐ Ctrl + U: Underline.
- ☐ Ctrl + E: Center-align text.
- ☐ Ctrl + L: Left-align text.
- ☐ Ctrl + R: Right-align text.

## ☐ **Document and Window Shortcuts:**

- ☐ Ctrl + N: Open a new document.
- ☐ Ctrl + O: Open an existing document.
- ☐ Ctrl + S: Save the document.
- ☐ Ctrl + P: Print the document.
- ☐ Ctrl + F4: Close the active window.
- ☐ Alt + F4: Close the entire program.

# MS Word Shortcut Keys

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## ☐ **Miscellaneous Shortcuts:**

- ☐ F7: Spell check.
- ☐ Ctrl + F: Find.
- ☐ Ctrl + H: Replace.
- ☐ Ctrl + K: Insert hyperlink.
- ☐ Ctrl + Space: Remove character formatting.
- ☐ Ctrl + 1, Ctrl + 2, Ctrl + 5: Single, double, or 1.5 line spacing.

## ☐ **Navigation Shortcuts:**

- ☐ Ctrl + Arrow Keys: Move the cursor by word.
- ☐ Ctrl + Backspace: Delete the previous word.
- ☐ Ctrl + Delete: Delete the next word.
- ☐ Ctrl + Home: Move to the beginning of the document.
- ☐ Ctrl + End: Move to the end of the document.
- ☐ Ctrl + Page Up: Move to the previous page.
- ☐ Ctrl + Page Down: Move to the next page.

# Any Question?

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# Thank you

# Information and Communication Technology

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Lecture : 14

Department of Computer Science  
GPGC SWABI (Khyber Pakhtunkhwa)

Fall 2025 Semester  
BS Course: ICT

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# MS PowerPoint

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- ❑ Microsoft PowerPoint, introduced in 1990, changed how people make and share presentations.
- ❑ It is used to create presentations for a various scenarios.
- ❑ It has an easy to use interface.
- ❑ PowerPoint works well with other apps like Word and Excel.
- ❑ You can save your presentations in different formats, like PDF, so it's easy to share them with others.

# MS PowerPoint (Versions)

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## ☐ **MS PowerPoint Versions:**

- ☐ PowerPoint 1.0 (1987)
- ☐ PowerPoint 2.0 (1988)
- ☐ PowerPoint 3.0 (1992)
- ☐ PowerPoint 4.0 (1994)
- ☐ PowerPoint 95 (7.0) (1995)
- ☐ PowerPoint 97 (8.0) (1997)
- ☐ PowerPoint 2000 (9.0) (1999)
- ☐ PowerPoint 2002 (10.0) (2001)
- ☐ PowerPoint 2003 (11.0) (2003)

- ☐ PowerPoint 2007 (12.0) (2007)
- ☐ PowerPoint 2010 (14.0) (2010)
- ☐ PowerPoint 2013 (15.0) (2013)
- ☐ PowerPoint 2016 (16.0) (2015)
- ☐ PowerPoint 2019 (16.0) (2018)
- ☐ PowerPoint 2022 (17.0) (2021)
- ☐ PowerPoint 365

# MS PowerPoint (Extensions)

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- ☐ The extension name of MS PowerPoint 2003 or earlier version is .ppt.
- ☐ The extension name of MS PowerPoint 2007 onwards all version are .pptx.

# MS PowerPoint (Features)

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- ☐ PowerPoint is easy to use, so anyone can create good-looking slideshows without being a design expert.
- ☐ It has lots of templates and themes to pick from, letting users make their presentations look the way they want.
- ☐ PowerPoint lets you add text, pictures, charts, and videos, making presentations interesting.
- ☐ It has cool features like animations and transitions that make presentations more exciting.
- ☐ You can put data from Excel and Word right into your PowerPoint presentation.
- ☐ PowerPoint lets many people work on a presentation at the same time, making it easy to collaborate.

# MS PowerPoint (How to open?)

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- ☐ How to open MS PowerPoint in Windows 10 and newer versions.

## 1) From the Start Menu:

- ☐ Click the Start Button at the left bottom of your screen to show the Start Menu.
- ☐ Scroll the all programs section.
- ☐ Choose MS PowerPoint.

## 2) From the Search bar:

- ☐ Click on the Search Bar in the taskbar and search for PowerPoint by typing PowerPoint.
- ☐ Click on MS PowerPoint.

# MS PowerPoint (How to open?)

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3) From right click menu:

- ☐ Right-click on any empty place
- ☐ Select New from the menu
- ☐ Then select MS PowerPoint Presentation from the list

# MS PowerPoint (New Presentation)

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- ☐ **Create a New Presentation**

- ☐ There are two ways to create new presentation:

1)

- ☐ Click the File Tab and Click New.
- ☐ Then click blank presentation or any other template as your like.

2)

- ☐ Press CTRL+N on the keyboard .
- ☐ It will open a new Word document.

- ☐ **Note:**

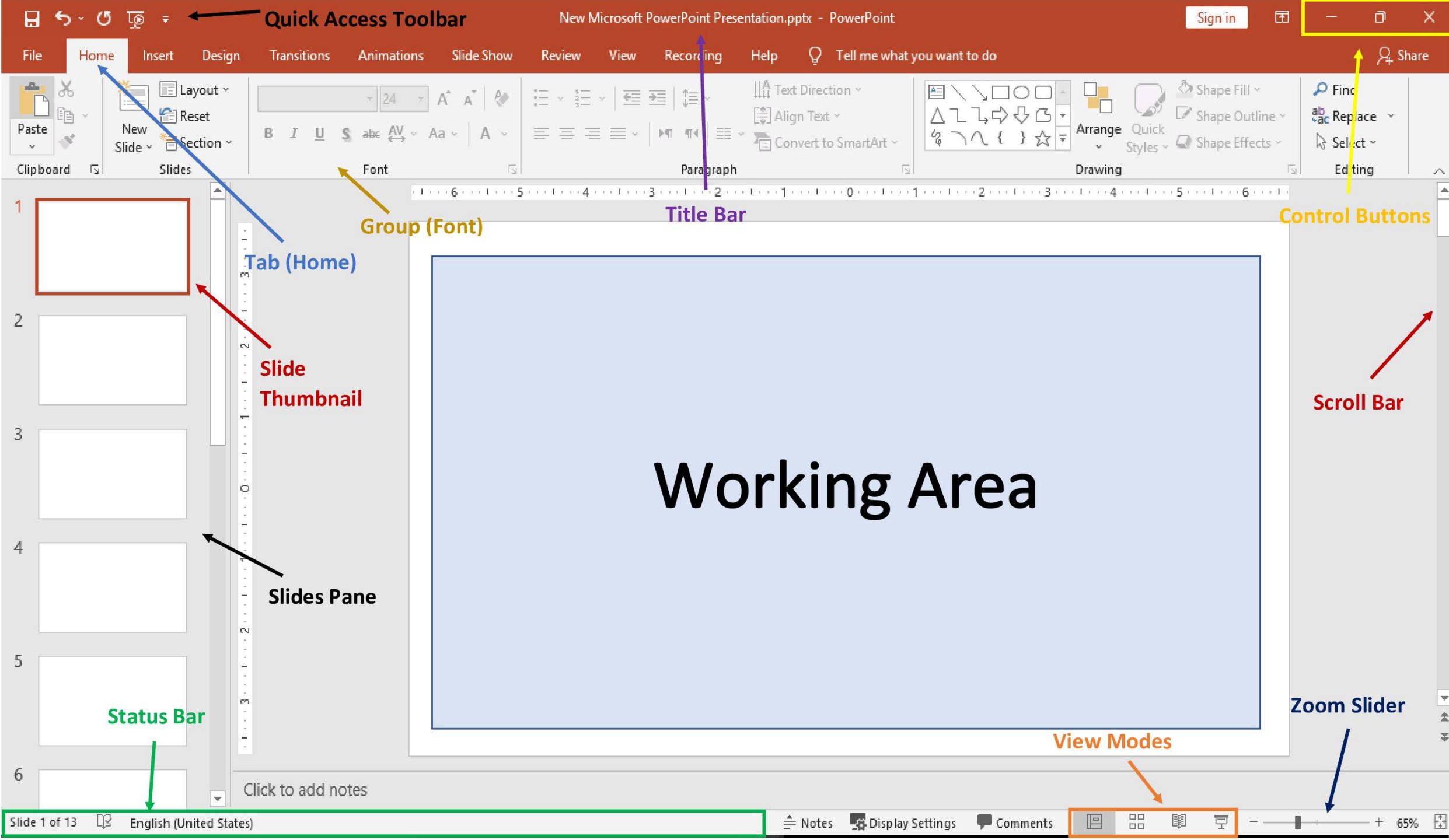
- ☐ When you click on the File tab and Click New, you have many choices about the types of presentation you can create. If you wish to start from a blank presentation, click Blank. If you wish to start from a template, you can browse through your choices on the screen.



# MS PowerPoint (Interface)

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- ❑ MS PowerPoint provides an easy to use interface to work with different types of presentations.



# MS PowerPoint (Interface)

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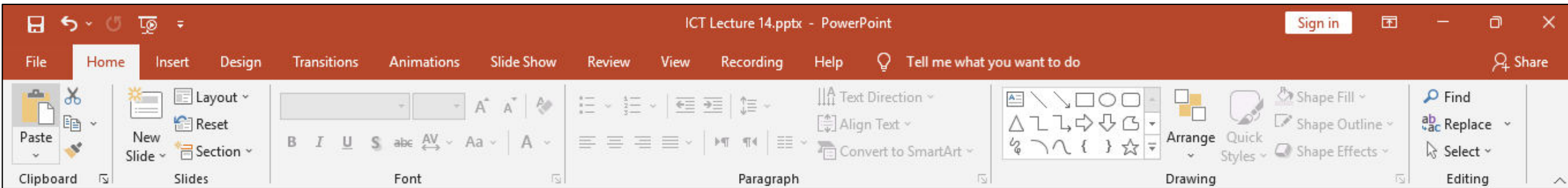
- ❑ **Quick Access Toolbar:**
- ❑ The quick access toolbar is a customizable toolbar that contains commands that you may want to use.
- ❑ You can place the quick access toolbar above or below the ribbon.
- ❑ You can also add items to the quick access toolbar.



# MS PowerPoint (Interface)

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- ❑ **The Ribbon:**
- ❑ The Ribbon is the panel at the top portion of the PowerPoint.
- ❑ It has nine tabs: File, Home, Insert, Design, Layout, References, Mailings, Review, and View that contain many new and existing features of PowerPoint.
- ❑ Each tab is divided into groups.



# MS PowerPoint (Interface)

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- ❑ **File:** Using file menu, one can save the file, close the file, open the file, create new document, print the file, customize the document, print, send (through email or fax), publish or close.
- ❑ **Home:** Clipboard, Fonts, Paragraph, Styles, and Editing.
- ❑ **Insert:** Pages, Tables, Illustrations, Links, Add-ins, Media, Comments, Header & Footer, Text, and Symbols
- ❑ **Design:** Themes, Page Background
- ❑ **Transitions:** Preview, Slide Transitions, Timing
- ❑ **Animations:** Preview, Animations, Advanced Animations, Timing

# MS PowerPoint (Interface)

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- ❑ **Slide Show:** Start Slide Show, Setup, Monitors
- ❑ **Review:** Proofing, Accessibility, Language, Comments, Compare,
- ❑ **View:** Presentation Views, Master Views, Show/Hide, Zoom, Window, Macros
- ❑ **Recording:** Record, Content, Auto-play Media, Save

# MS PowerPoint (Saving presentation)

## ❑ Saving a presentation:

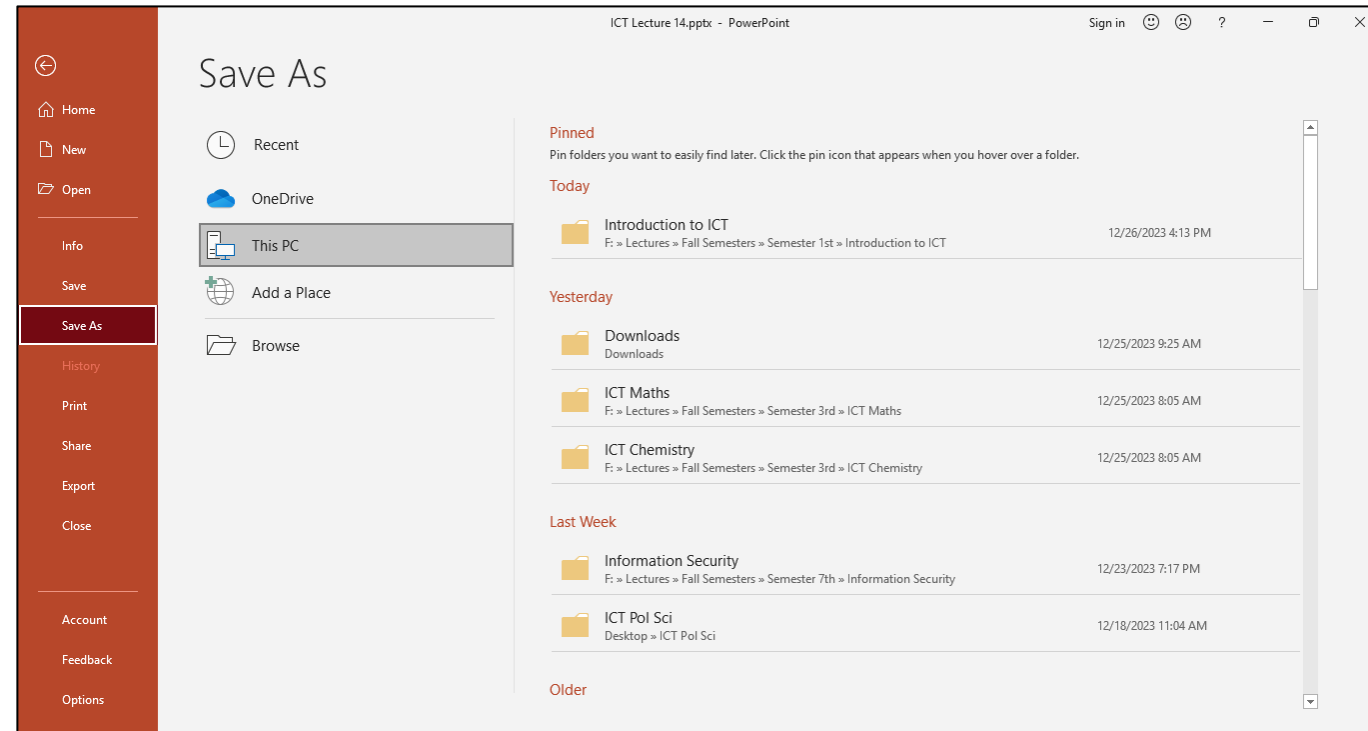
1) Click the File Tab and Click Save or Save As

OR

2) Click on save icon (towards the top-left on the screen)

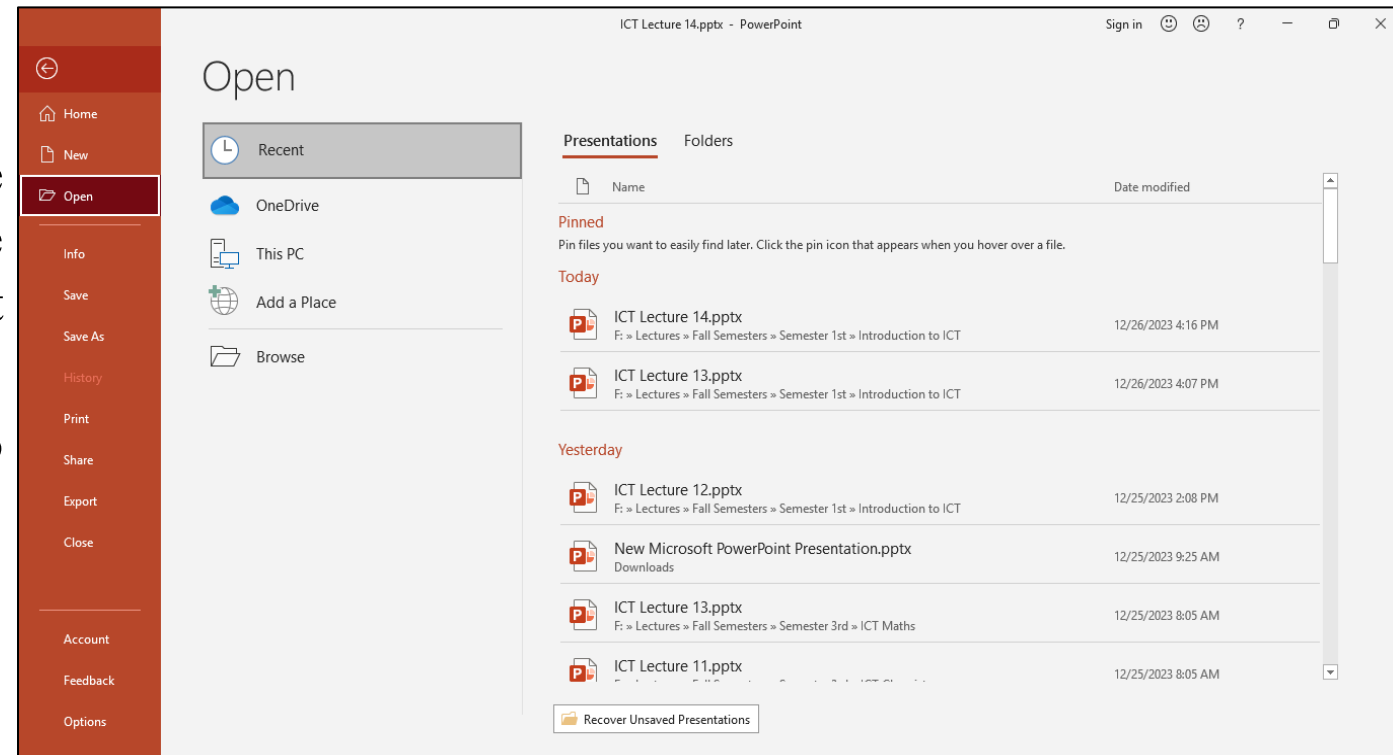
OR

3) Press CTRL+S on the keyboard and the file will be saved



# MS PowerPoint (Opening presentation)

- ❑ **Opening a presentation :**
- ❑ Click the File Tab and Click Open,
- ❑ If you have recently used the presentation, you can select and double click on presentation from the Right panel recent file list.
- ❑ Then select the file which you want to open from your saved location.





# MS PowerPoint (Opening presentation)

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OR

- ☐ If you want to open the file that is stored in your hard drive then
- ☐ Go to the PC and open the location where you have stored the file.
- ☐ Then select the file which you want to open from your saved location.
- ☐ Click open command button.

# MS PowerPoint (Renaming presentations)

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- ❑ **Renaming presentation:**
- ❑ To rename a presentation while using the program:
- ❑ Click the File tab and find the file you want to rename.
- ❑ Right-click the presentation name with the mouse and select Rename from the shortcut menu.
- ❑ Type the new name for the file and press the ENTER key.

# MS PowerPoint (Inserting text)

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- ☐ **Inserting Text, Images, etc.:**
- ☐ Text can be inserted in a presentation at any point using any of the following methods:
- ☐ **Type Text:**
- ☐ Put your cursor where you want to add the text and begin typing.

# MS PowerPoint (Cut, Copy, Paste)

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## ☐ **Copy and Paste Text:**

- ☐ Highlight the text you wish to copy and right click and click Copy, put your cursor where you want the text in the presentation and right click and click Paste.
- ☐ OR Ctrl + C for copying and Ctrl + V for pasting.

## ☐ **Cut and Paste Text:**

- ☐ Highlight the text you wish to copy, right click, and click Cut, put your cursor where you want the text in the presentation, right click, and click Paste.
- ☐ OR Ctrl + X for cutting and Ctrl + V for pasting.

# MS PowerPoint (Drag & Delete text)

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## ☐ **Drag Text:**

- ☐ Highlight the text you wish to move, click on it and drag it to the place where you want the text in the presentation.
- ☐ You can also use the Clipboard group on the Ribbon.

## ☐ **Deleting Blocks of Text:**

- ☐ Use the BACKSPACE and DELETE keys on the keyboard to delete text.
- ☐ Backspace will delete text to the left of the cursor and Delete will erase text to the right.
- ☐ To delete a large selection of text, highlight it using any of the methods outlined above and press the DELETE key.

# MS PowerPoint (Home Tab)

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## ☐ **Adding Slides:**

☐ You can insert new slides in the following ways:

1) Click the New Slide Button in the Slide group on the Home tab menu.

OR

2) Right click in the Slide Pane and Select New Slide.

OR

3) Click the New Slide Button in the Slide group on the Insert tab menu.

And select the required layout slide.

# MS PowerPoint (Home Tab)

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- ❑ **Font Group:**
- ❑ It allows formatting the text.
- ❑ It has some buttons such as bold, italic, underline, strikethrough, superscript, subscript, change case, font color, text highlight color, grow font button, shrink font button, font typeface and font size combo box.



# MS PowerPoint (Home Tab)

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- ❑ **Bold** button allows make the selected text bold.
- ❑ **Italic** button allows italicize the selected text.
- ❑ **Underline** button allows underline the selected text and also can change the underline style and color.
- ❑ **Strikethrough** allows draw a line through the middle of the selected text.
- ❑ **Change Case** buttons allows change all selected text to uppercase, lowercase or other common capitalizations.
- ❑ When you click on change case button, display some option such as Sentence case, lower case, UPPER CASE, Capitalize Each Word and Toggle Case.
- ❑ **Font color** list button allows change text color and it display some color.



# MS PowerPoint (Home Tab)

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- ☐ **To change the font type face:**
- ☐ First select the text which you want o changes.
- ☐ Click Home tab of the ribbon.
- ☐ Click the arrow next to the font name and choose a font of the font group of the ribbon.
- ☐ Remember that you can preview how the new font will look by highlighting the text, and hovering over the new font typeface.

# MS PowerPoint (Home Tab)

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- ☐ **To change the font size:**
- ☐ Click the arrow next to the font size and choose the appropriate size.
- OR
- ☐ Click the increase or decrease font size buttons.
- ☐ **Font Styles and Effects:**
- ☐ Font styles are predefined formatting options that are used to emphasize text.
- ☐ They include: Bold, Italic, and Underline. To add these to text:
- ☐ Select the text and click the Font Styles included on the Font Group of the Ribbon,
- OR
- ☐ Select the text and right click to display the font tools

# MS PowerPoint (Home Tab)

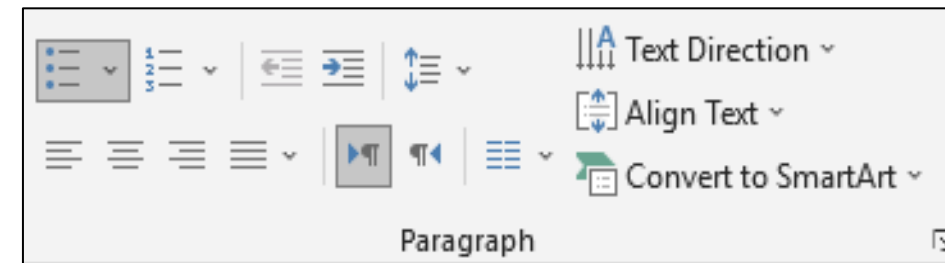
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- ☐ **Change Text Color:**
- ☐ Select the text and click the Colors button included in the Font Group of the Ribbon.  
OR
- ☐ Highlight the text, right click, and choose the colors tool.
- ☐ Select the color by clicking the down arrow next to the font color button.

# MS PowerPoint (Home Tab)

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- ❑ **Paragraph Group:**
- ❑ **Change Paragraph Alignment:**
- ❑ The paragraph alignment allows you to set how you want your text to appear.
- ❑ To change the alignment:
- ❑ Click the Home Tab
- ❑ Choose the appropriate button for alignment on the Paragraph Group.
  - ❑ **Align Left:** the text is aligned with your left margin
  - ❑ **Center:** The text is centered within your margins
  - ❑ **Align Right:** Aligns text with the right margin.
  - ❑ **Justify:** Aligns text to both the left and right margins.



# MS PowerPoint (Home Tab)

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- ❑ **Indent Paragraphs/Lists:**
- ❑ Indenting paragraphs/lists allows you set text within a paragraph at different margins.
- ❑ There are two buttons for setting indent level.
  - 1) Increase List Level
  - 2) Decrease List Level

# MS PowerPoint (Home Tab)

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- ☐ To indent paragraphs, you can do the following:
  - ☐ Click the Indent buttons to control the indent.
  - ☐ Click the Indent button repeated times to increase the size of the indent.
  - ☐ Click the dialog box of the Paragraph Group.
  - ☐ Click the Indents and Spacing Tab.
  - ☐ Select your indents.

# MS PowerPoint (Home Tab)

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- ☐ **Change Spacing Between Paragraphs and Lines:**
- ☐ You can change the space between lines and paragraphs by doing the following:
- ☐ Select the paragraph or paragraphs you wish to change.
- ☐ On the Home Tab, Click the Paragraph group launcher button
- ☐ In the Spacing section, adjust your spacing accordingly

# MS PowerPoint (Home Tab)

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- ☐ **Add fill and outline to Paragraphs and objects:**
- ☐ You can add fill and outlining to paragraphs and objects. To create a border around a paragraph or object:
- ☐ Select the text/object where you want the fill or outline.
- ☐ Click the fill/outline Button in the Drawing Group on the Home Tab
- ☐ Choose the appropriate options



# MS PowerPoint (Home Tab)

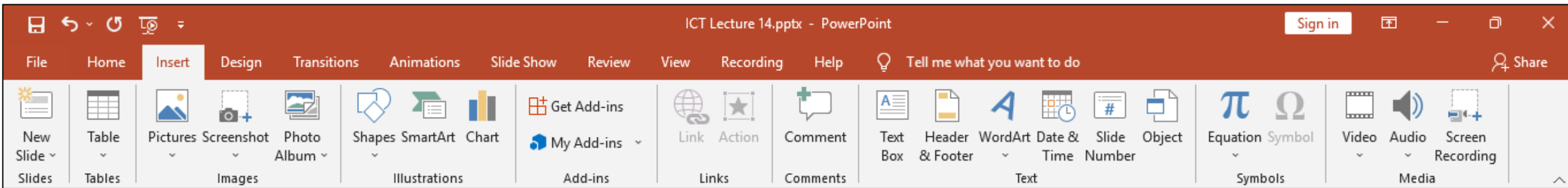
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- ☐ **Search and Replace Text:**
- ☐ To find a particular word or phrase in a presentation:
  - ☐ Click Home tab of the ribbon.
  - ☐ Click Find on the Editing Group.
- ☐ To find and replace a word or phrase in the presentation :
  - ☐ Click Home tab of the ribbon.
  - ☐ Click Replace on the Editing Group of the Ribbon. (Displays the find and replace dialog box)
- ☐ **Note:**
- ☐ Find and replace dialog box contain three tab such as find, replace and Go to.

# MS PowerPoint (Insert Tab)

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- ❑ By using insert Tab, we can insert specific object into current work area.
- ❑ Object means new slide, picture, Table, word art, Date & time, Any symbol, Chart, slide number, etc.



# MS PowerPoint (Insert Tab)

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- ❑ **Adding Tables:**
- ❑ Tables are used to display data in a table format.
- ❑ Place the cursor on the slide where you want the new slide
- ❑ Click the Insert Tab of the Ribbon
- ❑ Click the Table Button on the Tables Group. You can create a table one of four ways:
  - 1) Highlight the number of row and columns.
  - 2) Click Insert Table.
  - 3) Click the Draw Table, create your table by clicking and entering the rows and column.
  - 4) Click Excel Spreadsheet to insert a new spreadsheet table.

# MS PowerPoint (Insert Tab)

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- ❑ **Inserting Pictures:**
- ❑ PowerPoint allows you to insert pictures into a slide from online or your saved image.
- ❑ Place your cursor in the slide where you want the picture.
- ❑ Click the Insert Tab on the Ribbon.
- ❑ Click the picture button from the Images group.
- ❑ Select the picture from your local device or Online media.
- ❑ Click Insert.

# MS PowerPoint (Insert Tab)

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- ☐ **Inserting Illustrations and Smart Art:**
- ☐ PowerPoint allows you to insert illustrations and smart art into a document from online or your saved images.
- ☐ Place your cursor in the slide where you want the illustration/art,
- ☐ Click the Insert Tab on the Ribbon.
- ☐ Click the Shapes/SmartArt button from your illustrations groups.
- ☐ And select the required shape/art.

# MS PowerPoint (Insert Tab)

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- ☐ **Inserting Textbox:**
- ☐ To insert a text box in a slide.
  - ☐ Click the Insert tab on the ribbon.
  - ☐ Click the textbox button in the Text group
  - ☐ Now draw a textbox according to your requirements.

# MS PowerPoint (Insert Tab)

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- ❑ **Inserting Equations:**
- ❑ PowerPoint also allows you to insert mathematical equations.
- ❑ It allows to insert common mathematical equations or build up your own equations using a library of math symbols.
- ❑ To access the mathematical equations tool:
  - ❑ Place your cursor in the document where you want the equation
  - ❑ Click the Insert Tab on the Ribbon
  - ❑ Click the equations drop down Button.
  - ❑ Choose the appropriate equation and structure or click Insert New Equation
  - ❑ To edit the equation, click the equation and the Design Tab will be available in the Ribbon

# MS PowerPoint (Insert Tab)

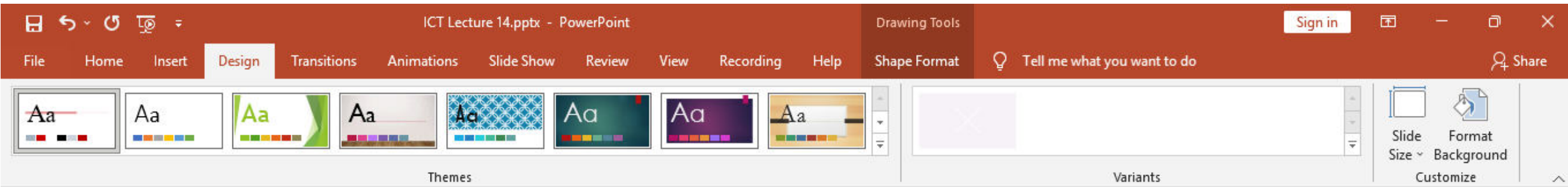
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- ☐ **Symbols and Special Characters:**
- ☐ Special characters are punctuation, spacing, or typographical characters that are not generally available on the standard keyboard. To insert symbols and special characters:
  - ☐ Place your cursor in the slide where you want the symbol
  - ☐ Click the Insert Tab on the Ribbon.
  - ☐ You will see the symbol button
  - ☐ Click the Symbols button.
  - ☐ It will display a table of symbols option.
  - ☐ Clicking on any one of these will insert the symbol where you clicked in the slide.
  - ☐ Then complete insert symbols, click close button.



# MS PowerPoint (Design Tab)

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- ☐ **Selecting themes:**
- ☐ Click the Design Tab on the Ribbon
- ☐ On the theme Group, select a theme that suits you.
- ☐ On clicking the More drop down button, a list will appear showing more themes.

# MS PowerPoint (Design Tab)

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- ☐ **Selecting Variants:**

- ☐ After selecting a theme:
- ☐ A list of variations appears in the Variants group.
- ☐ You can select your favorite variant from there.

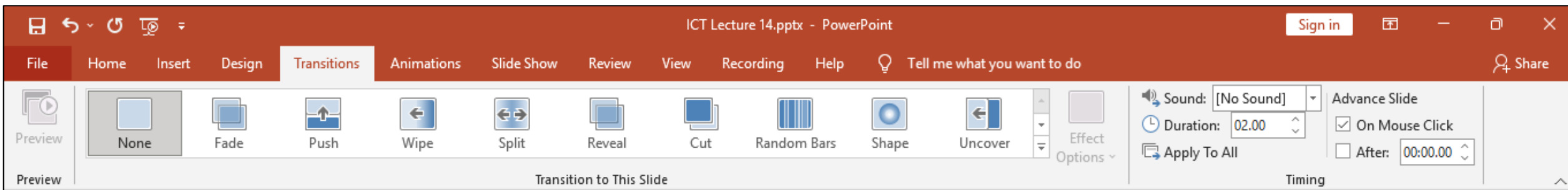
- ☐ **Customizing Slide:**

- ☐ You can customize your slide size and background using the Customize group

# MS PowerPoint (Transitions Tab)

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- ☐ The transition is an effect that occurs when two slides are switched.
- ☐ **Adding transitions to slides:**
- ☐ Click the slide on which you want to add a transition
- ☐ On the Transition Group, Click the transition you want.
- ☐ If you click the More button, a drop down list will be displayed showing more transition effects.
- ☐ Select the transition you want.



# MS PowerPoint (Transitions Tab)

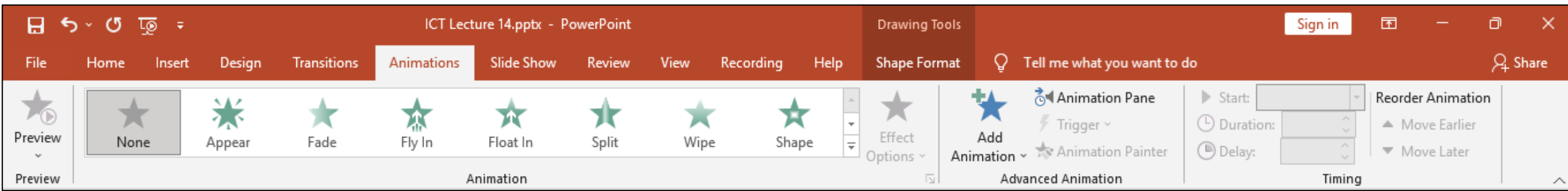
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- ☐ **Adding transition timing to slides:**
- ☐ First you add a transition effect to a slide.
- ☐ Then select the appropriate slide timings from the Timing group in the Transitions tab.

# MS PowerPoint (Animations Tab)

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- ☐ The animations is an effect on the object, when the object is introduced to the screen.
- ☐ **Adding animations to objects:**
- ☐ Click the object on which you want to add an animation
- ☐ On the Animation Group, Click the animation you want.
- ☐ If you click the More button, a drop down list will be displayed showing more transition effects.
- ☐ Select the animation you want.



# MS PowerPoint (Animations Tab)

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- ☐ **Adding advanced animations to objects:**
- ☐ Click the object on which you want to add an animation
- ☐ On the Advanced Animation Group, Click the add animation button.
- ☐ Through these options you can make your custom animations.

# MS PowerPoint (Animations Tab)

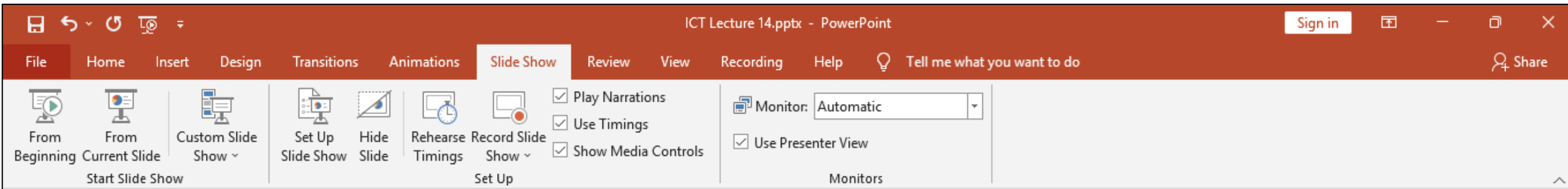
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- ☐ **Adding animation timing to slides:**
- ☐ First you add an animation effect to a slide.
- ☐ Then select the appropriate slide timings from the Timing group in the Animations tab.

# MS PowerPoint (Slide Show Tab)

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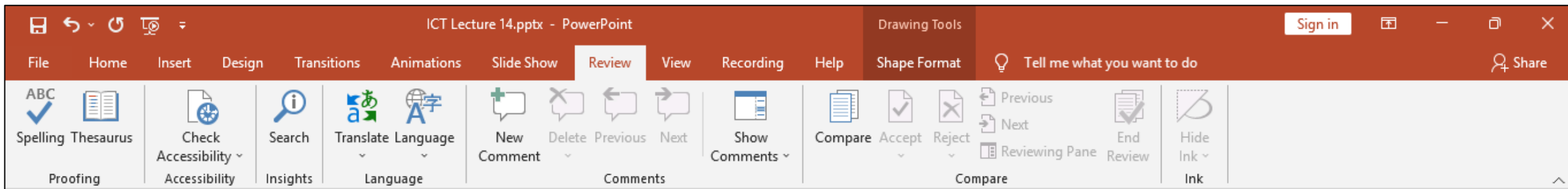
- ❑ This tab has options to start your slide from either:
  - ❑ Beginning of the Presentation
  - ❑ Current Slide
  - ❑ Create Custom Slide Show





# MS PowerPoint (Review Tab)

- ❑ The review tab includes three important features:
  - i. Spelling Check
  - ii. Thesaurus
  - iii. Language Translation



# MS PowerPoint (Review Tab)

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- ❑ **Spelling Check:**

- ❑ The spelling button on The Proofing group in the Review tab allow you to perform a spell check.

- ❑ **Thesaurus:**

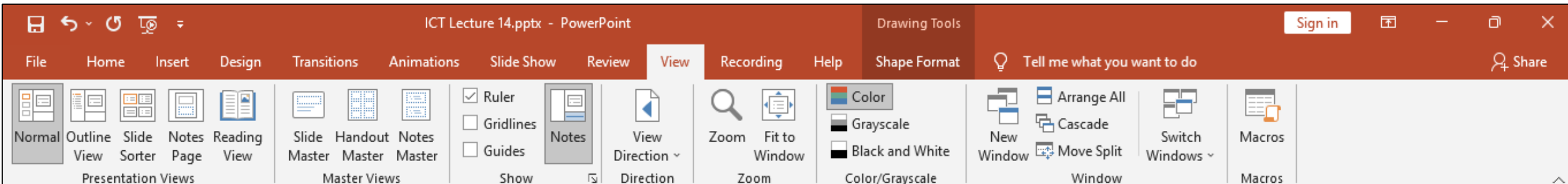
- ❑ It also include word suggestion tool, thesaurus, for word suggestion and replacement.

- ❑ **Language Selection/Translation:**

- ❑ It also includes features to select language or translate text to different languages.

# MS PowerPoint (View Tab)

- ❑ Microsoft PowerPoint offers several view modes to help users navigate and work on presentations in different ways. Here are the main view modes available in PowerPoint:



# MS PowerPoint (View Tab)

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## ☐ **Normal View:**

- ☐ This is the default view mode. It provides a workspace for designing and editing individual slides.

## ☐ **Outline View:**

- ☐ In Outline View, you can see the text content of your slides in a structured outline format.
- ☐ It's useful for quickly organizing and editing the textual content of your presentation.

## ☐ **Slide Sorter View:**

- ☐ Slide Sorter View displays a thumbnail grid of all the slides in your presentation.
- ☐ It allows you to easily rearrange, add, or delete slides, providing a high-level overview of your entire presentation.

# MS PowerPoint (View Tab)

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- ☐ **Notes Page View:**

- ☐ Notes Page View shows each slide along with any accompanying speaker notes. It's helpful for presenters who want to have reference notes during the presentation that are not visible to the audience.

- ☐ **Reading View:**

- ☐ Reading View is designed for rehearsing your presentation. It displays the slides in full screen without the editing interface, allowing you to practice your presentation in a more realistic setting.

# MS PowerPoint (View Tab)

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- ❑ **Slide Show:**
- ❑ Slide Show is not a traditional view within the PowerPoint interface but is a mode for presenting your slides. You can start the slide show from the beginning or from the current slide to preview how your presentation will look to the audience.
- ❑ To switch between these view modes in Microsoft PowerPoint, you can usually find options in the View tab on the Ribbon.

# MS PowerPoint (Printing)

## ☐ Printing Options:

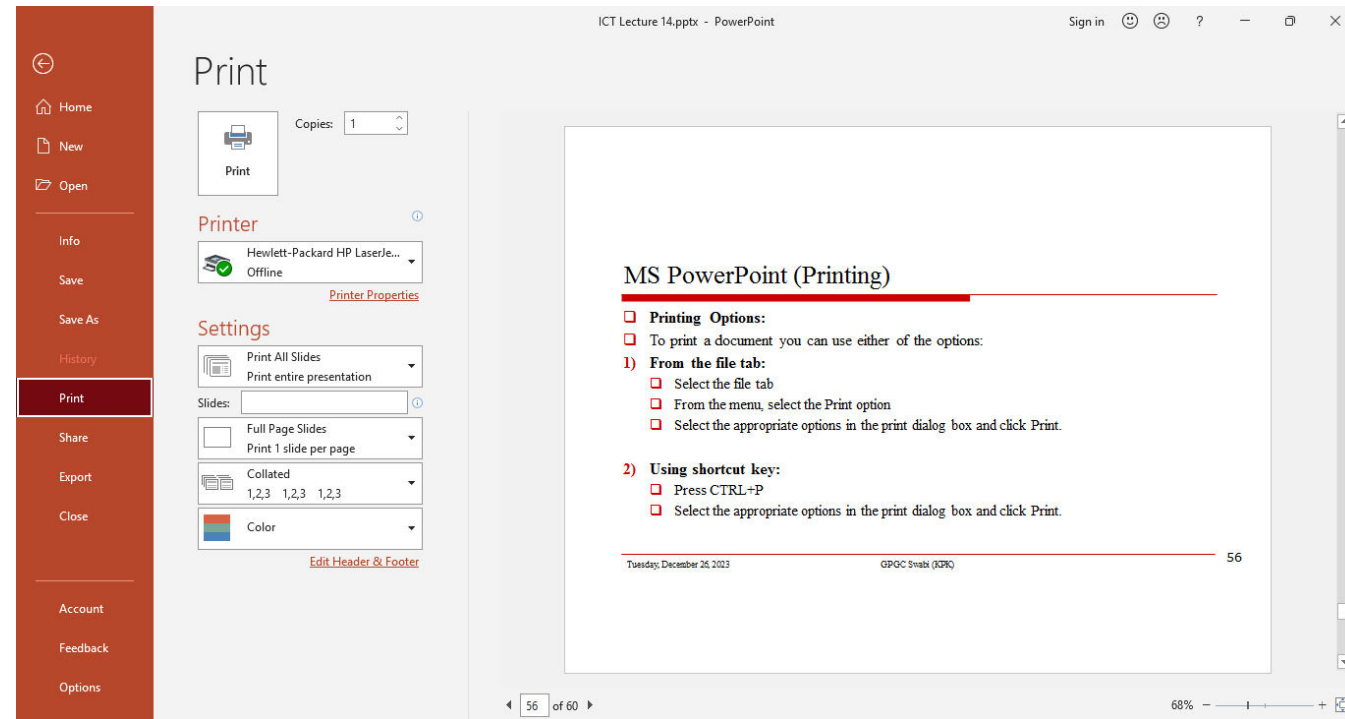
☐ To print a document you can use either of the options:

### 1) From the file tab:

- ☐ Select the file tab
- ☐ From the menu, select the Print option
- ☐ Select the appropriate options in the print dialog box and click Print.

### 2) Using shortcut key:

- ☐ Press CTRL+P
- ☐ Select the appropriate options in the print dialog box and click Print.



# MS PowerPoint Shortcut Keys

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## ☐ **General PowerPoint Shortcuts:**

- ☐ Ctrl + N: New presentation
- ☐ Ctrl + O: Open presentation
- ☐ Ctrl + S: Save presentation
- ☐ Ctrl + P: Print presentation
- ☐ Ctrl + Z: Undo
- ☐ Ctrl + Y: Redo
- ☐ Ctrl + X: Cut
- ☐ Ctrl + C: Copy
- ☐ Ctrl + V: Paste
- ☐ Ctrl + A: Select all



# MS PowerPoint Shortcut Keys

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## ☐ **Text Formatting Shortcuts:**

- ☐ Ctrl + B: Bold
- ☐ Ctrl + I: Italicize
- ☐ Ctrl + U: Underline
- ☐ Ctrl + E: Center text
- ☐ Ctrl + L: Left align text
- ☐ Ctrl + R: Right align text

## ☐ **Slide Navigation Shortcuts:**

- ☐ Ctrl + Home: Go to the beginning of the presentation
- ☐ Ctrl + End: Go to the end of the presentation
- ☐ Ctrl + N (in Slide Sorter View): Create a new slide
- ☐ Ctrl + Shift + N (in Normal View): Create a new slide

# MS PowerPoint Shortcut Keys

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## ☐ **Slide Editing Shortcuts:**

- ☐ Ctrl + D: Duplicate the selected object or slide
- ☐ Delete or Backspace: Delete the selected object or slide
- ☐ Ctrl + K: Insert hyperlink
- ☐ **Zooming Shortcuts:**
- ☐ Ctrl + Mouse Wheel: Zoom in or out in Normal View

## ☐ **Slide Show Shortcuts:**

- ☐ F5: Start the slide show from the beginning
- ☐ Shift + F5: Start the slide show from the current slide
- ☐ B or Period (.) key: Blank screen during a presentation
- ☐ Esc: End the slide show

# Any Question?

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# Thank you